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A Simple Framework for Contrastive Learning of Visual Representations

Ting Chen [1] **Simon Kornblith** [1] **Mohammad Norouzi** [1] **Geoffrey Hinton** [1]

Abstract

This paper presents *SimCLR*: a simple framework for contrastive learning of visual representations. We simplify recently proposed contrastive self-supervised learning algorithms without requiring specialized architectures or a memory bank. In order to understand what enables the contrastive prediction tasks to learn useful representations, we systematically study the major components of our framework. We show that (1) composition of data augmentations plays a critical role in defining effective predictive tasks, (2) introducing a learnable nonlinear transformation between the representation and the contrastive loss substantially improves the quality of the learned representations, and (3) contrastive learning benefits from larger batch sizes and more training steps compared to supervised learning. By combining these findings, we are able to considerably outperform previous methods for self-supervised and semi-supervised learning on ImageNet. A linear classifier trained on self-supervised representations learned by SimCLR achieves 76.5% top-1 accuracy, which is a 7% relative improvement over previous state-of-the-art, matching the performance of a supervised ResNet-50. When fine-tuned on only 1% of the labels, we achieve 85.8% top-5 accuracy, outperforming AlexNet with 100× fewer labels. [1]

1. Introduction

Learning effective visual representations without human supervision is a long-standing problem. Most mainstream approaches fall into one of two classes: generative or discriminative. Generative approaches learn to generate or otherwise model pixels in the input space (Hinton et al., 2006; Kingma & Welling, 2013; Goodfellow et al., 2014).

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[1]Code available at https://github.com/google-research/simclr.

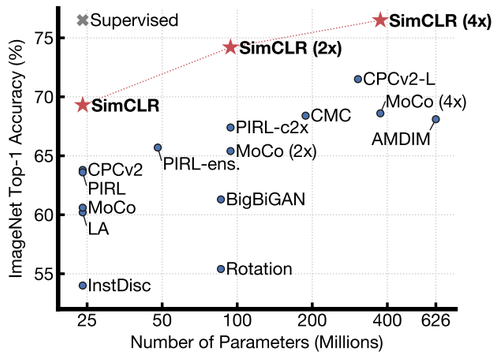


Figure 1. ImageNet Top-1 accuracy of linear classifiers trained on representations learned with different self-supervised methods (pretrained on ImageNet). Gray cross indicates supervised ResNet-50. Our method, SimCLR, is shown in bold.

However, pixel-level generation is computationally expensive and may not be necessary for representation learning. Discriminative approaches learn representations using objective functions similar to those used for supervised learning, but train networks to perform pretext tasks where both the inputs and labels are derived from an unlabeled dataset. Many such approaches have relied on heuristics to design pretext tasks (Doersch et al., 2015; Zhang et al., 2016; Noroozi & Favaro, 2016; Gidaris et al., 2018), which could limit the generality of the learned representations. Discriminative approaches based on contrastive learning in the latent space have recently shown great promise, achieving state-of-the-art results (Hadsell et al., 2006; Dosovitskiy et al., 2014; Oord et al., 2018; Bachman et al., 2019).

In this work, we introduce a simple framework for contrastive learning of visual representations, which we call *SimCLR*. Not only does SimCLR outperform previous work (Figure 1), but it is also simpler, requiring neither specialized architectures (Bachman et al., 2019; Hénaff et al., 2019) nor a memory bank (Wu et al., 2018; Tian et al., 2019; He et al., 2019; Misra & van der Maaten, 2019).

In order to understand what enables good contrastive representation learning, we systematically study the major components of our framework and show that:

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一个简单的 视觉表征对比学习的框架

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摘要

本文提出 SimCLR，一个用于视觉表征对比学习的简单框架。我们简化了近期的对比式自监督算法，不需要专门架构或记忆库。为理解对比预测任务如何学到有用表征，我们系统研究了框架中的主要组件。结果表明：（1）数据增强的组合，对定义有效预测任务至关重要；（2）在表征和对比损失之间加入可学习的非线性变换，显著提升表征质量；（3）相比监督学习，对比学习更能受益于更大批量和更多训练步数。结合这些发现，我们在 ImageNet 自监督与半监督学习上显著超过以往方法。在 SimCLR 学到的自监督表征上训练线性分类器，top-1 准确率达到 76.5%，相对此前最佳结果提高7%，与监督训练的 ResNet-50 相当。只用1%标签微调时，top-5 准确率达到85.8%，使用少 100× 的标签仍超过 AlexNet。 [1]

1 引言

在没有人工监督的情况下学习有效视觉表征，是长期存在的问题。主流方法通常分为生成式与判别式。生成式方法学习生成输入空间的像素，或对其进行建模（Hinton 等，2006；Kingma 与 Welling，2013；Goodfellow 等，2014）。

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第 37th 届国际机器学习会议论文集，奥地利维也纳，PMLR 119，2020。版权归作者所有。[1]代码：https://github.com/google-research/simclr。

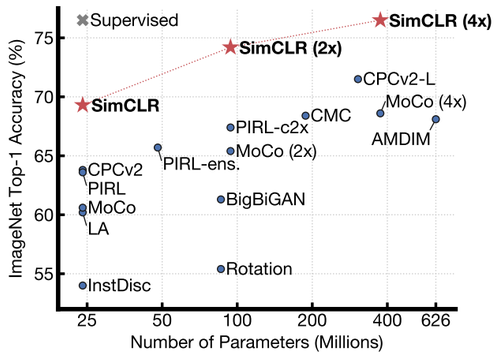


图1：在不同自监督方法学到的表征上训练线性分类器，所得 ImageNet top-1。均在 ImageNet 预训练。灰叉表示监督 ResNet-50，SimCLR 加粗显示。

然而，像素级生成的计算成本很高，而且表征学习未必需要它。判别式方法采用类似监督学习的目标函数，但训练网络完成代理任务，其输入和标签都来自无标签数据。许多方法依赖启发式规则设计代理任务（Doersch 等，2015；Zhang 等，2016；Noroozi 与 Favaro，2016；Gidaris 等，2018），这可能限制表征的通用性。近期，基于潜在空间对比学习的判别式方法显示出很大潜力，并取得了领先结果（Hadsell 等，2006；Dosovitskiy 等，2014；Oord 等，2018；Bachman 等，2019）。

本文提出用于视觉表征对比学习的简单框架 SimCLR。它不仅优于以往方法（图1），而且更简单：不需要专门架构（Bachman 等，2019；Hénaff 等，2019），也不需要记忆库（Wu 等，2018；Tian 等，2019；He 等，2019；Misra 与 van der Maaten，2019）。

为理解良好对比表征学习的关键，我们系统研究框架主要组件，发现：

- Composition of multiple data augmentation operations is crucial in defining the contrastive prediction tasks that yield effective representations. In addition, unsupervised contrastive learning benefits from stronger data augmentation than supervised learning.
- Introducing a learnable nonlinear transformation between the representation and the contrastive loss substantially improves the quality of the learned representations.
- Representation learning with contrastive cross entropy loss benefits from normalized embeddings and an appropriately adjusted temperature parameter.
- Contrastive learning benefits from larger batch sizes and longer training compared to its supervised counterpart. Like supervised learning, contrastive learning benefits from deeper and wider networks.

We combine these findings to achieve a new state-of-the-art in self-supervised and semi-supervised learning on ImageNet ILSVRC-2012 (Russakovsky et al., 2015). Under the linear evaluation protocol, SimCLR achieves 76.5% top-1 accuracy, which is a 7% relative improvement over previous state-of-the-art (Hénaff et al., 2019). When fine-tuned with only 1% of the ImageNet labels, SimCLR achieves 85.8% top-5 accuracy, a relative improvement of 10% (Hénaff et al., 2019). When fine-tuned on other natural image classification datasets, SimCLR performs on par with or better than a strong supervised baseline (Kornblith et al., 2019) on 10 out of 12 datasets.

2. Method

2.1. The Contrastive Learning Framework

Inspired by recent contrastive learning algorithms (see Section 7 for an overview), SimCLR learns representations by maximizing agreement between differently augmented views of the same data example via a contrastive loss in the latent space. As illustrated in Figure 2, this framework comprises the following four major components.

- A stochastic *data augmentation* module that transforms any given data example randomly resulting in two correlated views of the same example, denoted $\tilde{\boldsymbol{x}}_i$ and $\tilde{\boldsymbol{x}}_j$, which we consider as a positive pair. In this work, we sequentially apply three simple augmentations: *random cropping* followed by resize back to the original size, *random color distortions*, and *random Gaussian blur*. As shown in Section 3, the combination of random crop and color distortion is crucial to achieve a good performance.
- A neural network *base encoder* $f(\cdot)$ that extracts representation vectors from augmented data examples. Our framework allows various choices of the network architecture without any constraints. We opt for simplicity and adopt the commonly used ResNet (He et al., 2016)

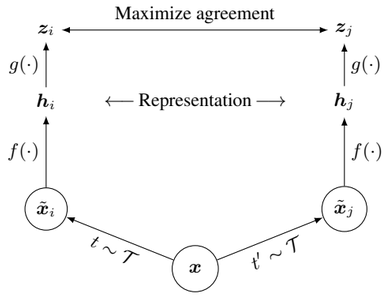


Figure 2. A simple framework for contrastive learning of visual representations. Two separate data augmentation operators are sampled from the same family of augmentations ($t \sim \mathcal{T}$ and $t' \sim \mathcal{T}$) and applied to each data example to obtain two correlated views. A base encoder network $f(\cdot)$ and a projection head $g(\cdot)$ are trained to maximize agreement using a contrastive loss. After training is completed, we throw away the projection head $g(\cdot)$ and use encoder $f(\cdot)$ and representation $\boldsymbol{h}$ for downstream tasks.

to obtain $\boldsymbol{h}_i = f(\tilde{\boldsymbol{x}}_i) = \text{ResNet}(\tilde{\boldsymbol{x}}_i)$ where $\boldsymbol{h}_i \in \mathbb{R}^d$ is the output after the average pooling layer.

- A small neural network *projection head* $g(\cdot)$ that maps representations to the space where contrastive loss is applied. We use a MLP with one hidden layer to obtain $\boldsymbol{z}_i = g(\boldsymbol{h}_i) = W^{(2)}\sigma(W^{(1)}\boldsymbol{h}_i)$ where σ is a ReLU nonlinearity. As shown in section 4, we find it beneficial to define the contrastive loss on $\boldsymbol{z}_i$'s rather than $\boldsymbol{h}_i$'s.
- A *contrastive loss function* defined for a contrastive prediction task. Given a set $\{\tilde{\boldsymbol{x}}_k\}$ including a positive pair of examples $\tilde{\boldsymbol{x}}_i$ and $\tilde{\boldsymbol{x}}_j$, the *contrastive prediction task* aims to identify $\tilde{\boldsymbol{x}}_j$ in $\{\tilde{\boldsymbol{x}}_k\}_{k\neq i}$ for a given $\tilde{\boldsymbol{x}}_i$.

We randomly sample a minibatch of N examples and define the contrastive prediction task on pairs of augmented examples derived from the minibatch, resulting in $2N$ data points. We do not sample negative examples explicitly. Instead, given a positive pair, similar to (Chen et al., 2017), we treat the other $2(N-1)$ augmented examples within a minibatch as negative examples. Let $\text{sim}(\boldsymbol{u}, \boldsymbol{v}) = \boldsymbol{u}^\top \boldsymbol{v}/\|\boldsymbol{u}\|\|\boldsymbol{v}\|$ denote the dot product between ℓ_2 normalized $\boldsymbol{u}$ and $\boldsymbol{v}$ (i.e. cosine similarity). Then the loss function for a positive pair of examples (i, j) is defined as

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} \exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_k)/\tau)}, \quad (1)$$

where $\mathbb{1}_{[k\neq i]} \in \{0, 1\}$ is an indicator function evaluating to 1 iff $k \neq i$ and τ denotes a temperature parameter. The final loss is computed across all positive pairs, both (i, j) and (j, i), in a mini-batch. This loss has been used in previous work (Sohn, 2016; Wu et al., 2018; Oord et al., 2018); for convenience, we term it *NT-Xent* (the normalized temperature-scaled cross entropy loss).

• 组合多种数据增强，对定义能产生有效表征的对比预测任务至关重要。此外，无监督对比学习比监督学习更能受益于更强的数据增强。

• 在表征和对比损失之间加入可学习的非线性变换，显著提高表征质量。

• 采用对比交叉熵的表征学习，受益于归一化嵌入及适当调整的温度。

• 相比监督学习，对比学习更能受益于大批量及长时间训练。与监督学习一样，对比学习也受益于更深、更宽的网络。

结合这些发现，我们在 ImageNet ILSVRC-2012（Russakovsky 等，2015）的自监督与半监督学习上取得新的领先结果。线性评估中，SimCLR 的 top-1 准确率达到76.5%，相对此前最佳结果提高7%（Hénaff 等，2019）。只使用 ImageNet 的1%标签微调时，top-5 准确率达到85.8%，相对提高10%。在其他自然图像分类数据集上微调时，12个数据集中有10个，SimCLR 与强监督基线（Kornblith 等，2019）相当或更好。

2 方法

2.1 对比学习框架

受近期对比学习算法启发，概览见第7节，SimCLR 通过潜在空间的对比损失，最大化同一样本不同增强视图之间的一致性，学习表征。图2展示了框架的四个主要组件。

• 随机数据增强模块，随机变换给定样本，得到同一样本的两个相关视图，记为 $\tilde{\boldsymbol{x}}_i$、$\tilde{\boldsymbol{x}}_j$，视为正样本对。本文依次采用三种简单增强：随机裁剪并缩放回原尺寸、随机颜色扰动、随机高斯模糊。第3节表明，裁剪与颜色扰动的组合对良好性能至关重要。

• 基础神经网络编码器 $f(\cdot)$，从增强样本提取表征向量。框架不限制网络架构，可采用多种选择。为保持简单，我们采用常用的 ResNet（He 等，2016）

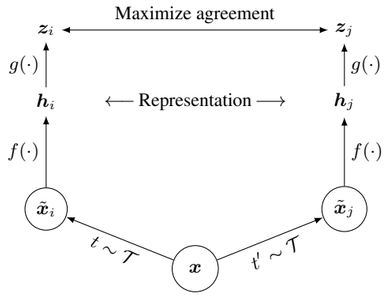


图2：视觉表征对比学习的简单框架。从同一增强族独立采样两个增强操作（$t \sim \mathcal{T}$、$t' \sim \mathcal{T}$），作用于每个样本，产生两个相关视图。训练基础编码器 $f(\cdot)$ 和投影头 $g(\cdot)$，通过对比损失最大化一致性。训练结束后，丢弃投影头 $g(\cdot)$，将编码器 $f(\cdot)$ 及表征 $\boldsymbol{h}$ 用于下游任务。

得到 $\boldsymbol{h}_i = f(\tilde{\boldsymbol{x}}_i)$ =ResNet（$\tilde{\boldsymbol{x}}_i$），其中 $\boldsymbol{h}_i \in \mathbb{R}^d$ 是平均池化层的输出。

• 小型神经网络投影头 $g(\cdot)$，把表征映射到计算对比损失的空间。我们使用含一个隐藏层的 MLP，得到 $\boldsymbol{z}_i = g(\boldsymbol{h}_i) = W(2)\sigma(W(1)\boldsymbol{h}_i)$，其中 σ 为 ReLU 非线性。第4节表明，在 $\boldsymbol{z}_i$ 而非 $\boldsymbol{h}_i$ 上定义对比损失，更有利。

• 为对比预测任务定义的对比损失。给定包含正样本对 $\tilde{\boldsymbol{x}}_i$、$\tilde{\boldsymbol{x}}_j$ 的集合 $\{\tilde{\boldsymbol{x}}_k\}$，对比预测任务的目标是：给定 $\tilde{\boldsymbol{x}}_i$，在 $\{\tilde{\boldsymbol{x}}_k\}_{k\neq i}$ 中识别 $\tilde{\boldsymbol{x}}_j$。

我们随机抽取包含 N 个样本的小批量，在由批内样本增强得到的配对样本上定义对比预测任务，总计 $2N$ 个数据点。不显式抽取负样本；对于一个正样本对，类似 Chen 等（2017），将批内其他2（$N-1$）个增强样本作为负样本。令 sim（$\boldsymbol{u}$，$\boldsymbol{v}$）$= \boldsymbol{u}^\top \boldsymbol{v}/\|\boldsymbol{u}\|\|\boldsymbol{v}\|$ 表示经 ℓ_2 归一化的 $\boldsymbol{u}$ 与 $\boldsymbol{v}$ 的点积，即余弦相似性。于是，正样本对（i，j）的损失定义为

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} \exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_k)/\tau)}, \quad (1)$$

其中 $\mathbb{1}_{[k\neq i]} \in \{0, 1\}$ 为指示函数，当且仅当 $k \neq i$ 时取1；τ 是温度参数。最终损失对批内全部正样本对计算，包含（i，j）及（j，i）两个方向。此前已有工作使用这一损失（Sohn，2016；Wu 等，2018；Oord 等，2018）。为方便，我们称其为 NT-Xent，即归一化、温度缩放的交叉熵损失。

Algorithm 1 SimCLR's main learning algorithm.

input: batch size N, constant τ, structure of $f, g, \mathcal{T}$.
for sampled minibatch $\{\boldsymbol{x}_k\}_{k=1}^{N}$ **do**
  for all $k \in \{1, \dots, N\}$ **do**
    draw two augmentation functions $t \sim \mathcal{T}$, $t' \sim \mathcal{T}$
    # the first augmentation
    $\tilde{\boldsymbol{x}}_{2k-1} = t(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k-1} = f(\tilde{\boldsymbol{x}}_{2k-1})$ # representation
    $\boldsymbol{z}_{2k-1} = g(\boldsymbol{h}_{2k-1})$ # projection
    # the second augmentation
    $\tilde{\boldsymbol{x}}_{2k} = t'(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k} = f(\tilde{\boldsymbol{x}}_{2k})$ # representation
    $\boldsymbol{z}_{2k} = g(\boldsymbol{h}_{2k})$ # projection
  end for
  for all $i \in \{1, \dots, 2N\}$ and $j \in \{1, \dots, 2N\}$ **do**
    $s_{i,j} = \boldsymbol{z}_i^\top \boldsymbol{z}_j / (\|\boldsymbol{z}_i\| \|\boldsymbol{z}_j\|)$ # pairwise similarity
  end for
  define $\ell(i, j)$ **as** $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$
  $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^{N} [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$
  update networks f and g to minimize $\mathcal{L}$
end for
return encoder network $f(\cdot)$, and throw away $g(\cdot)$

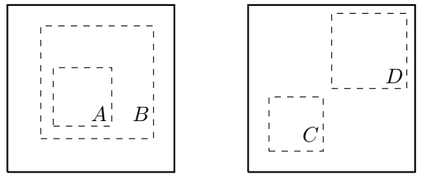
Algorithm 1 summarizes the proposed method.

2.2. Training with Large Batch Size

To keep it simple, we do not train the model with a memory bank (Wu et al., 2018; He et al., 2019). Instead, we vary the training batch size N from 256 to 8192. A batch size of 8192 gives us 16382 negative examples per positive pair from both augmentation views. Training with large batch size may be unstable when using standard SGD/Momentum with linear learning rate scaling (Goyal et al., 2017). To stabilize the training, we use the LARS optimizer (You et al., 2017) for all batch sizes. We train our model with Cloud TPUs, using 32 to 128 cores depending on the batch size.[2]

Global BN. Standard ResNets use batch normalization (Ioffe & Szegedy, 2015). In distributed training with data parallelism, the BN mean and variance are typically aggregated locally per device. In our contrastive learning, as positive pairs are computed in the same device, the model can exploit the local information leakage to improve prediction accuracy without improving representations. We address this issue by aggregating BN mean and variance over all devices during the training. Other approaches include shuffling data examples across devices (He et al., 2019), or replacing BN with layer norm (Hénaff et al., 2019).

[2]With 128 TPU v3 cores, it takes ∼1.5 hours to train our ResNet-50 with a batch size of 4096 for 100 epochs.



(a) Global and local views. (b) Adjacent views.

Figure 3. Solid rectangles are images, dashed rectangles are random crops. By randomly cropping images, we sample contrastive prediction tasks that include global to local view ($B \rightarrow A$) or adjacent view ($D \rightarrow C$) prediction.

2.3. Evaluation Protocol

Here we lay out the protocol for our empirical studies, which aim to understand different design choices in our framework.

Dataset and Metrics. Most of our study for unsupervised pretraining (learning encoder network f without labels) is done using the ImageNet ILSVRC-2012 dataset (Russakovsky et al., 2015). Some additional pretraining experiments on CIFAR-10 (Krizhevsky & Hinton, 2009) can be found in Appendix B.9. We also test the pretrained results on a wide range of datasets for transfer learning. To evaluate the learned representations, we follow the widely used linear evaluation protocol (Zhang et al., 2016; Oord et al., 2018; Bachman et al., 2019; Kolesnikov et al., 2019), where a linear classifier is trained on top of the frozen base network, and test accuracy is used as a proxy for representation quality. Beyond linear evaluation, we also compare against state-of-the-art on semi-supervised and transfer learning.

Default setting. Unless otherwise specified, for data augmentation we use random crop and resize (with random flip), color distortions, and Gaussian blur (for details, see Appendix A). We use ResNet-50 as the base encoder network, and a 2-layer MLP projection head to project the representation to a 128-dimensional latent space. As the loss, we use NT-Xent, optimized using LARS with learning rate of 4.8 ($= 0.3 \times \text{BatchSize}/256$) and weight decay of 10^{-6}. We train at batch size 4096 for 100 epochs.[3] Furthermore, we use linear warmup for the first 10 epochs, and decay the learning rate with the cosine decay schedule without restarts (Loshchilov & Hutter, 2016).

3. Data Augmentation for Contrastive Representation Learning

Data augmentation defines predictive tasks. While data augmentation has been widely used in both supervised and unsupervised representation learning (Krizhevsky et al.,

[3]Although max performance is not reached in 100 epochs, reasonable results are achieved, allowing fair and efficient ablations.

算法1：SimCLR 的主要学习算法。

输入：批量大小 N、常数 τ、 f、 g、 $\mathcal{T}$ 的结构。
对采样的小批量 $\{\boldsymbol{x}_k\}_{k=1}^{N}$ **do**
  对所有 $k \in \{1, \dots, N\}$ **执行**
    抽取两个增强函数 $t \sim \mathcal{T}$、 $t' \sim \mathcal{T}$
    # 第一次增强
    $\tilde{\boldsymbol{x}}_{2k-1} = t(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k-1} = f(\tilde{\boldsymbol{x}}_{2k-1})$ # 表征
    $\boldsymbol{z}_{2k-1} = g(\boldsymbol{h}_{2k-1})$ # 投影
    # 第二次增强
    $\tilde{\boldsymbol{x}}_{2k} = t'(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k} = f(\tilde{\boldsymbol{x}}_{2k})$ # 表征
    $\boldsymbol{z}_{2k} = g(\boldsymbol{h}_{2k})$ # 投影
  循环结束
  对全部 $i \in \{1, \dots, 2N\}$ **和** $j \in \{1, \dots, 2N\}$ **执行**
    $s_{i,j} = \boldsymbol{z}_i^\top \boldsymbol{z}_j / (\|\boldsymbol{z}_i\| \|\boldsymbol{z}_j\|)$ # 两两相似性
  循环结束
  将 $\ell(i, j)$ **定义为** $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$
  $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^{N} [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$
  更新网络 f 和 g，最小化 $\mathcal{L}$
循环结束
返回编码器 $f(\cdot)$，丢弃 $g(\cdot)$

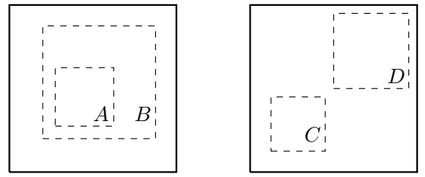
算法1总结了所提出的方法。

2.2 大批量训练

为保持简单，我们不使用记忆库训练（Wu 等，2018；He 等，2019），而将批量大小 N 从256变到8192。批量8192时，对每个正样本对，两种增强视图合计提供16382个负样本。使用标准 SGD/动量优化器和线性学习率缩放（Goyal 等，2017）时，大批量训练可能不稳定，因此所有批量都使用 LARS。训练在云 TPU 上进行，依批量大小使用32至128个核心。[2]

全局批归一化。标准 ResNet 使用批归一化（BN；Ioffe 与 Szegedy，2015）。数据并行的分布式训练通常在各设备内分别汇总 BN 均值和方差。我们的正样本对在同一设备计算，因此模型可能利用局部信息泄漏提高预测准确率，却不改善表征。我们通过在全部设备之间汇总 BN 统计量解决这一问题。其他解决方式包括跨设备打乱样本（He 等，2019），或用层归一化替换 BN（Hénaff 等，2019）。

[2]使用128个 TPUv3 核心，ResNet-50 以4096批量训练100轮，需要 ∼1.5小时。



(a) Global and local views. (b) Adjacent views.

图3：实线矩形表示图像，虚线矩形表示随机裁剪。通过随机裁剪，我们采样对比预测任务，包含全局到局部视图（$B \rightarrow A$）或相邻视图（$D \rightarrow C$）预测。

2.3 评估协议

这里介绍实证研究协议，目标是理解框架中的不同设计选择。

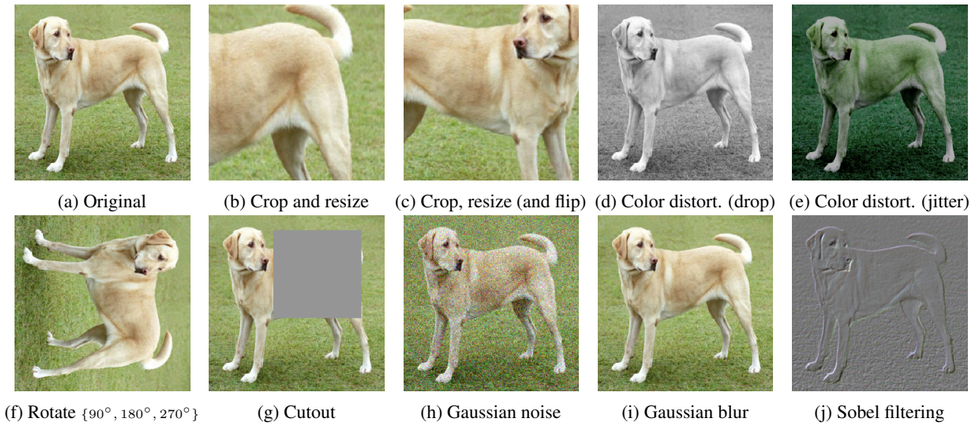
数据集与指标。无监督预训练，即在不使用标签的情况下学习编码器 f，主要在 ImageNet ILSVRC-2012（Russakovsky 等，2015）上进行。CIFAR-10 的补充预训练实验见附录B.9。我们也在多种数据集上评估迁移效果。评估表征时，使用广泛采用的线性评估协议（Zhang 等，2016；Oord 等，2018；Bachman 等，2019；Kolesnikov 等，2019）：冻结基础网络，在其上训练线性分类器，以测试准确率作为表征质量的代理指标。除了线性评估，我们还在半监督与迁移学习设置中，与领先方法比较。

默认设置。除非另有说明，数据增强使用随机裁剪缩放与翻转、颜色扰动及高斯模糊，详见附录A。基础编码器为 ResNet-50；两层 MLP 投影头将表征映射到128维潜在空间。损失为 NT-Xent，使用 LARS 优化，学习率4.8（$= 0.3 \times \text{BatchSize}/256$），权重衰减 10^{-6}。批量大小4096，训练100轮。[3] 前10轮采用线性预热，之后使用不重启的余弦学习率衰减（Loshchilov 与 Hutter，2016）。

3 对比表征学习的数据增强

数据增强定义预测任务。虽然增强已广泛用于监督和无监督表征学习（Krizhevsky 等，

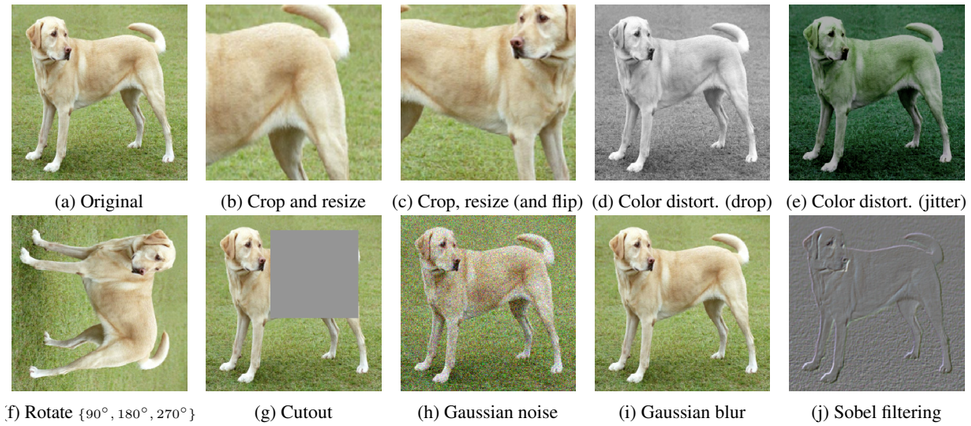
[3]虽然100轮尚未达到最大性能，但已取得合理结果，能够进行公平、高效的消融。



(a) Original (b) Crop and resize (c) Crop, resize (and flip) (d) Color distort. (drop) (e) Color distort. (jitter)

(f) Rotate $\{90°, 180°, 270°\}$ (g) Cutout (h) Gaussian noise (i) Gaussian blur (j) Sobel filtering

Figure 4. Illustrations of the studied data augmentation operators. Each augmentation can transform data stochastically with some internal parameters (e.g. rotation degree, noise level). Note that we *only* test these operators in ablation, the *augmentation policy used to train our models* only includes *random crop (with flip and resize)*, *color distortion*, and *Gaussian blur*. (Original image cc-by: Von.grzanka)



(a) Original (b) Crop and resize (c) Crop, resize (and flip) (d) Color distort. (drop) (e) Color distort. (jitter)

(f) Rotate $\{90°, 180°, 270°\}$ (g) Cutout (h) Gaussian noise (i) Gaussian blur (j) Sobel filtering

图4：所研究增强操作的示意。各增强都可通过内部参数，如旋转角度、噪声强度，随机变换数据。请注意，这些操作只在消融中测试，实际训练策略仅含随机裁剪、翻转缩放、颜色扰动及高斯模糊。原图署名 cc-by：Von.grzanka。

2012; Hénaff et al., 2019; Bachman et al., 2019), it has not been considered as a systematic way to define the contrastive prediction task. Many existing approaches define contrastive prediction tasks by changing the architecture. For example, Hjelm et al. (2018); Bachman et al. (2019) achieve global-to-local view prediction via constraining the receptive field in the network architecture, whereas Oord et al. (2018); Hénaff et al. (2019) achieve neighboring view prediction via a fixed image splitting procedure and a context aggregation network. We show that this complexity can be avoided by performing simple *random cropping* (with resizing) of target images, which creates a family of predictive tasks subsuming the above mentioned two, as shown in Figure 3. This simple design choice conveniently decouples the predictive task from other components such as the neural network architecture. Broader contrastive prediction tasks can be defined by extending the family of augmentations and composing them stochastically.

3.1. Composition of data augmentation operations is crucial for learning good representations

To systematically study the impact of data augmentation, we consider several common augmentations here. One type of augmentation involves spatial/geometric transformation of data, such as cropping and resizing (with horizontal flipping), rotation (Gidaris et al., 2018) and cutout (DeVries & Taylor, 2017). The other type of augmentation involves appearance transformation, such as color distortion (including color dropping, brightness, contrast, saturation, hue) (Howard, 2013; Szegedy et al., 2015), Gaussian blur, and Sobel filtering. Figure 4 visualizes the augmentations that we study in this work.

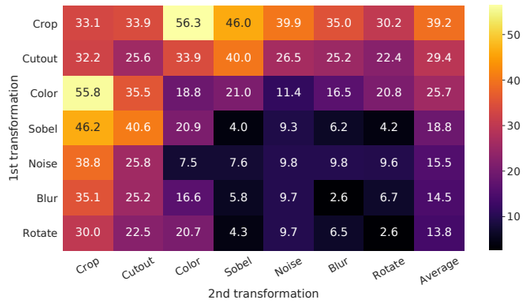


Figure 5. Linear evaluation (ImageNet top-1 accuracy) under individual or composition of data augmentations, applied only to one branch. For all columns but the last, diagonal entries correspond to single transformation, and off-diagonals correspond to composition of two transformations (applied sequentially). The last column reflects the average over the row.

To understand the effects of individual data augmentations and the importance of augmentation composition, we investigate the performance of our framework when applying augmentations individually or in pairs. Since ImageNet images are of different sizes, we always apply crop and resize images (Krizhevsky et al., 2012; Szegedy et al., 2015), which makes it difficult to study other augmentations in the absence of cropping. To eliminate this confound, we consider an asymmetric data transformation setting for this ablation. Specifically, we always first randomly crop images and resize them to the same resolution, and we then apply the targeted transformation(s) *only* to one branch of the framework in Figure 2, while leaving the other branch as the identity (i.e. $t(\boldsymbol{x}_i) = \boldsymbol{x}_i$). Note that this asymmet-

2012；Hénaff 等，2019；Bachman 等，2019），但尚未被视为系统定义对比预测任务的方法。许多现有方法通过改变架构定义任务。例如，Hjelm 等（2018）和 Bachman 等（2019）通过限制网络感受野，实现全局到局部视图的预测；Oord 等（2018）和 Hénaff 等（2019）则通过固定图像分割与上下文聚合网络，预测相邻视图。我们表明，只需对目标图像随机裁剪并缩放，即可避免这种复杂性，并构造包含上述两类任务的一组预测任务，如图3所示。这一简单选择将预测任务与神经网络架构等其他组件分离。扩展增强种类并随机组合，还能定义更广泛的对比预测任务。

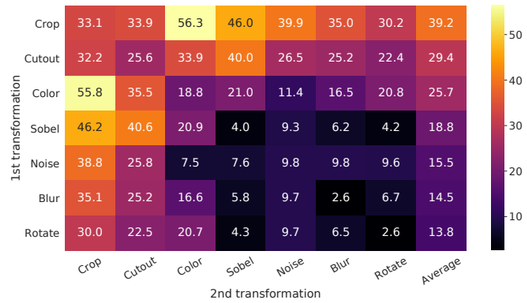
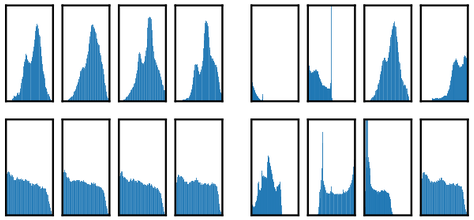


图5：仅对一条分支施加单种或组合增强后的线性评估，指标为 ImageNet top-1。除最后一列外，对角项表示单种变换，非对角项表示按顺序施加两种变换；最后一列为该行平均值。

3.1 数据增强组合对学习良好表征至关重要

为系统研究数据增强的影响，我们考察若干常见增强。一类涉及空间或几何变换，如裁剪缩放与水平翻转、旋转（Gidaris 等，2018）和区域遮挡 cutout（DeVries 与 Taylor，2017）。另一类改变外观，如颜色扰动，包括去色、亮度、对比度、饱和度、色相变化（Howard，2013；Szegedy 等，2015），以及高斯模糊和 Sobel 滤波。图4展示本文研究的增强。

为理解单种增强的作用及增强组合的重要性，我们分别使用单种或两种增强研究框架性能。由于 ImageNet 图像尺寸不同，我们始终进行裁剪和缩放（Krizhevsky 等，2012；Szegedy 等，2015），这使得在完全不裁剪时研究其他增强很困难。为消除这一混杂因素，消融采用非对称变换：先随机裁剪并缩放到相同分辨率，再只对图2的一条分支施加待研究变换，另一条分支使用恒等映射，即 $t(\boldsymbol{x}_i) = \boldsymbol{x}_i$。请注意，这种非对称



(a) Without color distortion. (b) With color distortion.

Figure 6. Histograms of pixel intensities (over all channels) for different crops of two different images (i.e. two rows). The image for the first row is from Figure 4. All axes have the same range.

Methods	Color distortion strength 1/8	1/4	1/2	1	1 (+Blur)	AutoAug
SimCLR	59.6	61.0	62.6	63.2	64.5	61.1
Supervised	77.0	76.7	76.5	75.7	75.4	77.1

Table 1. Top-1 accuracy of unsupervised ResNet-50 using linear evaluation and supervised ResNet-50[5], under varied color distortion strength (see Appendix A) and other data transformations. Strength 1 (+Blur) is our default data augmentation policy.

ric data augmentation hurts the performance. Nonetheless, this setup should not substantively change the impact of individual data augmentations or their compositions.

Figure 5 shows linear evaluation results under individual and composition of transformations. We observe that *no single transformation suffices to learn good representations*, even though the model can almost perfectly identify the positive pairs in the contrastive task. When composing augmentations, the contrastive prediction task becomes harder, but the quality of representation improves dramatically. Appendix B.2 provides a further study on composing broader set of augmentations.

One composition of augmentations stands out: random cropping and random color distortion. We conjecture that one serious issue when using only random cropping as data augmentation is that most patches from an image share a similar color distribution. Figure 6 shows that color histograms alone suffice to distinguish images. Neural nets may exploit this shortcut to solve the predictive task. Therefore, it is critical to compose cropping with color distortion in order to learn generalizable features.

3.2. Contrastive learning needs stronger data augmentation than supervised learning

To further demonstrate the importance of the color augmentation, we adjust the strength of color augmentation as

[5]Supervised models are trained for 90 epochs; longer training improves performance of stronger augmentation by $\sim 0.5\%$.

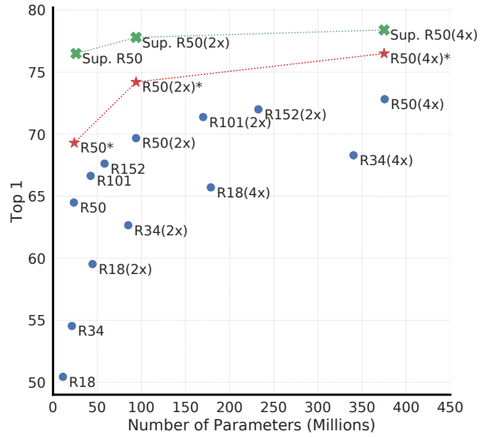


Figure 7. Linear evaluation of models with varied depth and width. Models in blue dots are ours trained for 100 epochs, models in red stars are ours trained for 1000 epochs, and models in green crosses are supervised ResNets trained for 90 epochs[7] (He et al., 2016).

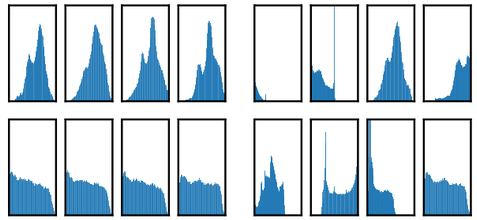
shown in Table 1. Stronger color augmentation substantially improves the linear evaluation of the learned unsupervised models. In this context, AutoAugment (Cubuk et al., 2019), a sophisticated augmentation policy found using supervised learning, does not work better than simple cropping + (stronger) color distortion. When training supervised models with the same set of augmentations, we observe that stronger color augmentation does not improve or even hurts their performance. Thus, our experiments show that unsupervised contrastive learning benefits from stronger (color) data augmentation than supervised learning. Although previous work has reported that data augmentation is useful for self-supervised learning (Doersch et al., 2015; Bachman et al., 2019; Hénaff et al., 2019; Asano et al., 2019), we show that data augmentation that does not yield accuracy benefits for supervised learning can still help considerably with contrastive learning.

4. Architectures for Encoder and Head

4.1. Unsupervised contrastive learning benefits (more) from bigger models

Figure 7 shows, perhaps unsurprisingly, that increasing depth and width both improve performance. While similar findings hold for supervised learning (He et al., 2016), we find the gap between supervised models and linear classifiers trained on unsupervised models shrinks as the model size increases, suggesting that *unsupervised learning benefits more from bigger models than its supervised counterpart*.

[7]Training longer does not improve supervised ResNets (see Appendix B.3).



(a) Without color distortion. (b) With color distortion.

图6：两张不同图像，各对应一行，其不同裁剪的像素强度直方图，合并全部通道。第一行图像来自图4。所有坐标轴范围相同。

方法	颜色扰动强度 1/8	1/4	1/2	1	1（+模糊）	AutoAug
SimCLR	59.6	61.0	62.6	63.2	64.5	61.1
监督	77.0	76.7	76.5	75.7	75.4	77.1

表1：不同颜色扰动强度，见附录A，及其他变换下，无监督 ResNet-50 的线性评估 top-1 与监督 ResNet-50[5]，的准确率。强度1（+模糊）为默认增强策略。

增强会损害性能。不过，这一设置不应实质改变单种增强及其组合的影响。

图5展示单种及组合变换的线性评估。虽然模型在对比任务中几乎能完美识别正样本对，但单一变换都不足以学到好表征。组合增强使对比预测更难，却显著改善表征质量。附录B.2进一步研究更广泛的增强组合。

随机裁剪与随机颜色扰动的组合尤其突出。我们猜测，只用裁剪的一个严重问题是：同一图像的多数图像块具有相近颜色分布。图6显示，单靠颜色直方图就足以区分图像。网络可能利用这一捷径完成预测，因此将裁剪与颜色扰动组合，对学习可泛化特征至关重要。

3.2 对比学习需要比监督学习更强的数据增强

为进一步说明颜色增强的重要性，我们按以下方式调整增强强度：

[5]监督模型训练90轮；延长训练可使更强增强的性能提高 ~ 0.5 %。

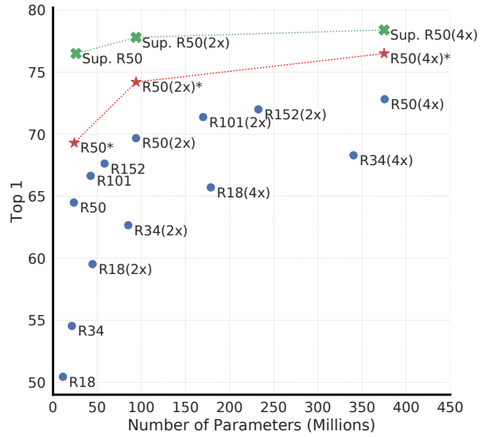


图7：不同深度和宽度模型的线性评估。蓝点为我们训练100轮的模型，红星为训练1000轮，绿叉为监督训练90轮的 ResNet[7]（He 等，2016）。

结果见表1。更强的颜色增强显著改善了无监督模型的线性评估。在这里，用监督学习搜索得到的复杂增强策略 AutoAugment（Cubuk 等，2019），并不优于简单裁剪+更强的颜色扰动。使用相同增强训练监督模型时，我们发现，更强的颜色增强没有改善性能，甚至会使性能下降。因此，无监督对比学习比监督学习更能受益于较强的颜色增强。虽然已有工作指出数据增强有益于自监督学习（Doersch 等，2015；Bachman 等，2019；Hénaff 等，2019；Asano 等，2019），但我们进一步表明：即使某种增强不能提升监督学习准确率，它仍可能显著帮助对比学习。

4 编码器与投影头的架构

4.1 无监督对比学习更能受益于大型模型

图7表明，增加深度和宽度都会提高性能，这并不意外。监督学习也有类似发现（He 等，2016），但我们发现，模型越大，监督模型与在无监督表征上训练的线性分类器之间的差距越小，说明无监督学习比监督学习更能受益于大型模型。

[7]延长训练不能改善监督 ResNet，见附录B.3。

Name	Negative loss function	Gradient w.r.t. $\boldsymbol{u}$
NT-Xent	$\boldsymbol{u}^T\boldsymbol{v}^+/\tau - \log \sum_{\boldsymbol{v}\in\{\boldsymbol{v}^+,\boldsymbol{v}^-\}} \exp(\boldsymbol{u}^T\boldsymbol{v}/\tau)$	$(1 - \frac{\exp(\boldsymbol{u}^T\boldsymbol{v}^+/\tau)}{Z(\boldsymbol{u})})/\tau\boldsymbol{v}^+ - \sum_{\boldsymbol{v}^-} \frac{\exp(\boldsymbol{u}^T\boldsymbol{v}^-/\tau)}{Z(\boldsymbol{u})}/\tau\boldsymbol{v}^-$
NT-Logistic	$\log \sigma(\boldsymbol{u}^T\boldsymbol{v}^+/\tau) + \log \sigma(-\boldsymbol{u}^T\boldsymbol{v}^-/\tau)$	$(\sigma(-\boldsymbol{u}^T\boldsymbol{v}^+/\tau))/\tau\boldsymbol{v}^+ - \sigma(\boldsymbol{u}^T\boldsymbol{v}^-/\tau)/\tau\boldsymbol{v}^-$
Margin Triplet	$-\max(\boldsymbol{u}^T\boldsymbol{v}^- - \boldsymbol{u}^T\boldsymbol{v}^+ + m, 0)$	$\boldsymbol{v}^+ - \boldsymbol{v}^-$ if $\boldsymbol{u}^T\boldsymbol{v}^+ - \boldsymbol{u}^T\boldsymbol{v}^- < m$ else $\boldsymbol{0}$

Table 2. Negative loss functions and their gradients. All input vectors, i.e. $\boldsymbol{u}, \boldsymbol{v}^+, \boldsymbol{v}^-$, are ℓ_2 normalized. NT-Xent is an abbreviation for "Normalized Temperature-scaled Cross Entropy". Different loss functions impose different weightings of positive and negative examples.

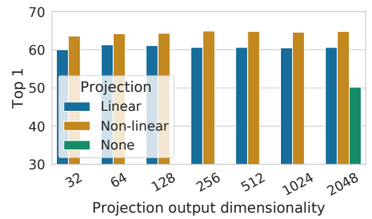


Figure 8. Linear evaluation of representations with different projection heads $g(\cdot)$ and various dimensions of $\boldsymbol{z} = g(\boldsymbol{h})$. The representation $\boldsymbol{h}$ (before projection) is 2048-dimensional here.

4.2. A nonlinear projection head improves the representation quality of the layer before it

We then study the importance of including a projection head, i.e. $g(\boldsymbol{h})$. Figure 8 shows linear evaluation results using three different architecture for the head: (1) identity mapping; (2) linear projection, as used by several previous approaches (Wu et al., 2018); and (3) the default nonlinear projection with one additional hidden layer (and ReLU activation), similar to Bachman et al. (2019). We observe that a nonlinear projection is better than a linear projection (+3%), and much better than no projection (>10%). When a projection head is used, similar results are observed regardless of output dimension. Furthermore, even when nonlinear projection is used, the layer before the projection head, $\boldsymbol{h}$, is still much better (>10%) than the layer after, $\boldsymbol{z} = g(\boldsymbol{h})$, which shows that *the hidden layer before the projection head is a better representation than the layer after*.

We conjecture that the importance of using the representation before the nonlinear projection is due to loss of information induced by the contrastive loss. In particular, $\boldsymbol{z} = g(\boldsymbol{h})$ is trained to be invariant to data transformation. Thus, g can remove information that may be useful for the downstream task, such as the color or orientation of objects. By leveraging the nonlinear transformation $g(\cdot)$, more information can be formed and maintained in $\boldsymbol{h}$. To verify this hypothesis, we conduct experiments that use either $\boldsymbol{h}$ or $g(\boldsymbol{h})$ to learn to predict the transformation applied during the pretraining. Here we set $g(h) = W^{(2)}\sigma(W^{(1)}h)$, with the same input and output dimensionality (i.e. 2048). Table 3 shows $\boldsymbol{h}$ contains much more information about the transformation applied, while $g(\boldsymbol{h})$ loses information. Further analysis can be found in Appendix B.4.

What to predict?	Random guess	Representation $\boldsymbol{h}$	Representation $g(\boldsymbol{h})$
Color vs grayscale	80	99.3	97.4
Rotation	25	67.6	25.6
Orig. vs corrupted	50	99.5	59.6
Orig. vs Sobel filtered	50	96.6	56.3

Table 3. Accuracy of training additional MLPs on different representations to predict the transformation applied. Other than crop and color augmentation, we additionally and independently add rotation (one of $\{0°, 90°, 180°, 270°\}$), Gaussian noise, and Sobel filtering transformation during the pretraining for the last three rows. Both $\boldsymbol{h}$ and $g(\boldsymbol{h})$ are of the same dimensionality, i.e. 2048.

5. Loss Functions and Batch Size

5.1. Normalized cross entropy loss with adjustable temperature works better than alternatives

We compare the NT-Xent loss against other commonly used contrastive loss functions, such as logistic loss (Mikolov et al., 2013), and margin loss (Schroff et al., 2015). Table 2 shows the objective function as well as the gradient to the input of the loss function. Looking at the gradient, we observe 1) ℓ_2 normalization (i.e. cosine similarity) along with temperature effectively weights different examples, and an appropriate temperature can help the model learn from hard negatives; and 2) unlike cross-entropy, other objective functions do not weigh the negatives by their relative hardness. As a result, one must apply semi-hard negative mining (Schroff et al., 2015) for these loss functions: instead of computing the gradient over all loss terms, one can compute the gradient using semi-hard negative terms (*i.e.*, those that are within the loss margin and closest in distance, but farther than positive examples).

To make the comparisons fair, we use the same ℓ_2 normalization for all loss functions, and we tune the hyperparameters, and report their best results.[8] Table 4 shows that, while (semi-hard) negative mining helps, the best result is still much worse than our default NT-Xent loss.

[8]Details can be found in Appendix B.10. For simplicity, we only consider the negatives from one augmentation view.

Name	负损失函数	关于 $\boldsymbol{u}$ 的梯度
NT-Xent	$\boldsymbol{u}^T\boldsymbol{v}^+/\tau - \log \sum_{\boldsymbol{v}\in\{\boldsymbol{v}^+,\boldsymbol{v}^-\}} \exp(\boldsymbol{u}^T\boldsymbol{v}/\tau)$	$(1 - \frac{\exp(\boldsymbol{u}^T\boldsymbol{v}^+/\tau)}{Z(\boldsymbol{u})})/\tau\boldsymbol{v}^+ - \sum_{\boldsymbol{v}^-} \frac{\exp(\boldsymbol{u}^T\boldsymbol{v}^-/\tau)}{Z(\boldsymbol{u})}/\tau\boldsymbol{v}^-$
NT-Logistic	$\log \sigma(\boldsymbol{u}^T\boldsymbol{v}^+/\tau) + \log \sigma(-\boldsymbol{u}^T\boldsymbol{v}^-/\tau)$	$(\sigma(-\boldsymbol{u}^T\boldsymbol{v}^+/\tau))/\tau\boldsymbol{v}^+ - \sigma(\boldsymbol{u}^T\boldsymbol{v}^-/\tau)/\tau\boldsymbol{v}^-$
间隔三元组损失	$-\max(\boldsymbol{u}^T\boldsymbol{v}^- - \boldsymbol{u}^T\boldsymbol{v}^+ + m, 0)$	若 $\boldsymbol{u}^T\boldsymbol{v}^+ - \boldsymbol{u}^T\boldsymbol{v}^- < m$，取 $\boldsymbol{v}^+ - \boldsymbol{v}^-$；否则取0

表2：负损失函数及其梯度。全部输入向量 $\boldsymbol{u}$、$\boldsymbol{v}^+$、$\boldsymbol{v}^-$ 均经 ℓ_2 归一化。NT-Xent 是"归一化、温度缩放的交叉熵"的缩写。不同损失对正、负样本赋予不同权重。

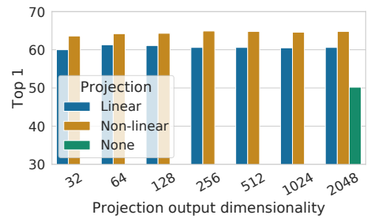


图8：不同投影头 $g(\cdot)$ 及不同 $\boldsymbol{z} = g(\boldsymbol{h})$ 维度下的线性评估。这里投影前表征 $\boldsymbol{h}$ 为2048维。

4.2 非线性投影头改善其前一层的表征质量

随后研究投影头的重要性，即 $g(\boldsymbol{h})$。图8展示三种投影头架构的线性评估结果：（1）恒等映射；（2）以往多种方法采用的线性投影（Wu 等，2018）；（3）默认的非线性投影，额外含一个使用 ReLU 的隐藏层，类似 Bachman 等（2019）。非线性投影优于线性投影（+3%），也远优于不使用投影（>10%）。使用投影头时，不同输出维度的结果相近。而且，即便使用非线性投影，投影头前的 $\boldsymbol{h}$ 仍显著优于其后的 $\boldsymbol{z} = g(\boldsymbol{h})$，差距为 >10%。这说明投影头前的隐藏层比投影头后更适合作为表征。

我们猜测，非线性投影前的表征之所以重要，是因为对比损失会造成信息丢失。具体而言，$\boldsymbol{z} = g(\boldsymbol{h})$ 被训练为对数据变换保持不变，因此 g 可能丢弃对下游任务有用的信息，例如物体的颜色或方向。借助非线性变换 $g(\cdot)$，可以在 $\boldsymbol{h}$ 中形成并保留更多信息。为验证这一假设，我们分别用 $\boldsymbol{h}$ 或 $g(\boldsymbol{h})$ 预测预训练时施加的变换。这里设 $g(h) = W^{(2)}\sigma(W^{(1)}h)$，输入输出维度相同，均为2048。表3表明，$\boldsymbol{h}$ 包含更多变换信息，而 $g(\boldsymbol{h})$ 丢失了信息。进一步分析可见附录B.4。

预测什么？	随机猜测	表征 $\boldsymbol{h}$	表征 $g(\boldsymbol{h})$
颜色与灰度	80	99.3	97.4
旋转	25	67.6	25.6
原图与损坏图像	50	99.5	59.6
原图与 Sobel 滤波图	50	96.6	56.3

表3：在不同表征上训练额外 MLP，预测所施加变换的准确率。除裁剪和颜色增强外，最后三行还分别在预训练中独立加入旋转，即 $\{0°, 90°, 180°, 270°\}$ 之一、高斯噪声及 Sobel 滤波。$\boldsymbol{h}$ 与 $g(\boldsymbol{h})$ 维度相同，均为2048。

5 损失函数与批量大小

5.1 可调温度的归一化交叉熵，优于其他选择

我们将 NT-Xent 与常用的对比损失比较，包括逻辑损失（Mikolov 等，2013）和间隔损失（Schroff 等，2015）。表2展示目标函数及其对损失输入的梯度。观察梯度发现：（1）ℓ_2 归一化，即余弦相似性，与温度共同对不同样本加权；合适的温度有助于模型从困难负样本中学习；（2）与交叉熵不同，其他目标不按相对难度给负样本加权。因此，这些损失需要半困难负样本挖掘（Schroff 等，2015）：不对所有损失项计算梯度，而只使用距离落在损失间隔内、且最接近但仍远于正样本的负样本项。

为公平比较，所有损失均使用相同的 ℓ_2 归一化，分别优化超参数，并报告最佳结果。[8] 表4表明，虽然半困难负样本挖掘有帮助，其最佳结果仍明显低于默认 NT-Xent。

[8]详见附录B.10。为简化，只考虑一种增强视图的负样本。

Margin	NT-Logi.	Margin (sh)	NT-Logi.(sh)	NT-Xent
50.9	51.6	57.5	57.9	63.9

Table 4. Linear evaluation (top-1) for models trained with different loss functions. "sh" means using semi-hard negative mining.

ℓ_2 norm?	τ	Entropy	Contrastive acc.	Top 1
Yes	0.05	1.0	90.5	59.7
	0.1	4.5	87.8	64.4
	0.5	8.2	68.2	60.7
	1	8.3	59.1	58.0
No	10	0.5	91.7	57.2
	100	0.5	92.1	57.0

Table 5. Linear evaluation for models trained with different choices of ℓ_2 norm and temperature τ for NT-Xent loss. The contrastive distribution is over 4096 examples.

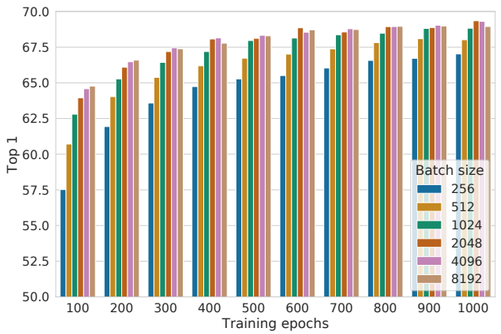


Figure 9. Linear evaluation models (ResNet-50) trained with different batch size and epochs. Each bar is a single run from scratch.[10]

We next test the importance of the ℓ_2 normalization (i.e. cosine similarity vs dot product) and temperature τ in our default NT-Xent loss. Table 5 shows that without normalization and proper temperature scaling, performance is significantly worse. Without ℓ_2 normalization, the contrastive task accuracy is higher, but the resulting representation is worse under linear evaluation.

5.2. Contrastive learning benefits (more) from larger batch sizes and longer training

Figure 9 shows the impact of batch size when models are trained for different numbers of epochs. We find that, when the number of training epochs is small (e.g. 100 epochs), larger batch sizes have a significant advantage over the smaller ones. With more training steps/epochs, the gaps between different batch sizes decrease or disappear, provided the batches are randomly resampled. In contrast to supervised learning (Goyal et al., 2017), in contrastive learning, larger batch sizes provide more negative examples, facilitating convergence (i.e. taking fewer epochs and steps for a given accuracy). Training longer also provides more negative examples, improving the results. In Appendix B.1, results with even longer training steps are provided.

[10] A linear learning rate scaling is used here. Figure B.1 shows using a square root learning rate scaling can improve performance of ones with small batch sizes.

Method	Architecture	Param (M)	Top 1	Top 5
Methods using ResNet-50:				
Local Agg.	ResNet-50	24	60.2	-
MoCo	ResNet-50	24	60.6	-
PIRL	ResNet-50	24	63.6	-
CPC v2	ResNet-50	24	63.8	85.3
SimCLR (ours)	ResNet-50	24	**69.3**	**89.0**
Methods using other architectures:				
Rotation	RevNet-50 (4×)	86	55.4	-
BigBiGAN	RevNet-50 (4×)	86	61.3	81.9
AMDIM	Custom-ResNet	626	68.1	-
CMC	ResNet-50 (2×)	188	68.4	88.2
MoCo	ResNet-50 (4×)	375	68.6	-
CPC v2	ResNet-161 (∗)	305	71.5	90.1
SimCLR (ours)	ResNet-50 (2×)	94	74.2	92.0
SimCLR (ours)	ResNet-50 (4×)	375	**76.5**	**93.2**

Table 6. ImageNet accuracies of linear classifiers trained on representations learned with different self-supervised methods.

Method	Architecture	Label fraction 1%	10%
		Top 5	
Supervised baseline	ResNet-50	48.4	80.4
Methods using other label-propagation:			
Pseudo-label	ResNet-50	51.6	82.4
VAT+Entropy Min.	ResNet-50	47.0	83.4
UDA (w. RandAug)	ResNet-50	-	88.5
FixMatch (w. RandAug)	ResNet-50	-	89.1
S4L (Rot+VAT+En. M.)	ResNet-50 (4×)	-	91.2
Methods using representation learning only:			
InstDisc	ResNet-50	39.2	77.4
BigBiGAN	RevNet-50 (4×)	55.2	78.8
PIRL	ResNet-50	57.2	83.8
CPC v2	ResNet-161(∗)	77.9	91.2
SimCLR (ours)	ResNet-50	75.5	87.8
SimCLR (ours)	ResNet-50 (2×)	83.0	91.2
SimCLR (ours)	ResNet-50 (4×)	**85.8**	**92.6**

Table 7. ImageNet accuracy of models trained with few labels.

6. Comparison with State-of-the-art

In this subsection, similar to Kolesnikov et al. (2019); He et al. (2019), we use ResNet-50 in 3 different hidden layer widths (width multipliers of 1×, 2×, and 4×). For better convergence, our models here are trained for 1000 epochs.

Linear evaluation. Table 6 compares our results with previous approaches (Zhuang et al., 2019; He et al., 2019; Misra & van der Maaten, 2019; Hénaff et al., 2019; Kolesnikov et al., 2019; Donahue & Simonyan, 2019; Bachman et al.,

间隔	NT-Logi.	间隔损失（sh）	NT-Logi.(sh)	NT-Xent
50.9	51.6	57.5	57.9	63.9

表4：使用不同损失训练模型的线性评估 top-1。"sh"表示半困难负样本挖掘。

使用ℓ_2归一化?	τ	熵	对比任务准确率	Top 1
Yes	0.05	1.0	90.5	59.7
	0.1	4.5	87.8	64.4
	0.5	8.2	68.2	60.7
	1	8.3	59.1	58.0
No	10	0.5	91.7	57.2
	100	0.5	92.1	57.0

表5：NT-Xent 使用不同 ℓ_2 范数及温度 τ 时的线性评估。对比分布包含4096个样本。

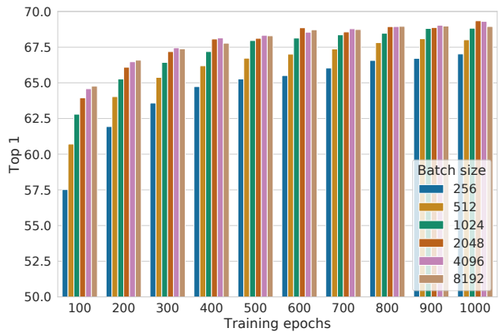


图9：不同批量、轮数训练的 ResNet-50 的线性评估。每根柱为一次从头训练。[10]

随后研究默认 NT-Xent 中 ℓ_2 归一化，即余弦相似性相对于点积，以及温度 τ 的重要性。表5表明，没有归一化和适当温度缩放，性能显著更差。没有 ℓ_2 归一化时，对比任务准确率更高，但线性评估表明所得表征更差。

5.2 对比学习更能受益于大批量与更长训练

图9展示不同训练轮数下批量大小的影响。训练轮数较少，如100轮时，大批量明显优于小批量。更多训练步或轮数下，只要批次随机重新采样，不同批量之间的差距会缩小或消失。相比之下，

[10] 这里采用线性学习率缩放。图B.1表明，平方根缩放能改善小批量性能。

方法	架构	参数量（百万）	Top-1	Top 5
使用 *ResNet-50* 的方法：				
局部聚合	ResNet-50	24	60.2	-
MoCo	ResNet-50	24	60.6	-
PIRL	ResNet-50	24	63.6	-
CPC v2	ResNet-50	24	63.8	85.3
SimCLR（本文）	ResNet-50	24	**69.3**	**89.0**
使用其他架构的方法：				
Rotation	RevNet-50 (4×)	86	55.4	-
BigBiGAN	RevNet-50 (4×)	86	61.3	81.9
AMDIM	自定义 ResNet	626	68.1	-
CMC	ResNet-50 (2×)	188	68.4	88.2
MoCo	ResNet-50 (4×)	375	68.6	-
CPC v2	ResNet-161 (∗)	305	71.5	90.1
SimCLR（本文）	ResNet-50 (2×)	94	74.2	92.0
SimCLR（本文）	ResNet-50 (4×)	375	**76.5**	**93.2**

表6：不同自监督方法所得表征上，训练线性分类器的 ImageNet 准确率。

方法	架构	标签比例 1%	10%
		Top 5	
监督基线	ResNet-50	48.4	80.4
使用其他标签传播的方法：			
伪标签	ResNet-50	51.6	82.4
VAT+熵最小化	ResNet-50	47.0	83.4
UDA (w. RandAug)	ResNet-50	-	88.5
FixMatch (w. RandAug)	ResNet-50	-	89.1
S4L (Rot+VAT+En. M.)	ResNet-50 (4×)	-	91.2
只使用表征学习的方法：			
InstDisc	ResNet-50	39.2	77.4
BigBiGAN	RevNet-50 (4×)	55.2	78.8
PIRL	ResNet-50	57.2	83.8
CPC v2	ResNet-161(∗)	77.9	91.2
SimCLR（本文）	ResNet-50	75.5	87.8
SimCLR（本文）	ResNet-50 (2×)	83.0	91.2
SimCLR（本文）	ResNet-50 (4×)	**85.8**	**92.6**

表7：少量标签训练模型的 ImageNet 准确率。

监督学习（Goyal 等，2017），对比学习中的更大批量提供更多负样本，帮助收敛，即达到指定准确率所需轮数和步数更少。延长训练同样提供更多负样本，从而改善结果。更长训练的结果见附录B.1。

6 与最先进方法比较

本节类似 Kolesnikov 等（2019）和 He 等（2019），使用三种隐藏层宽度的 ResNet-50，宽度倍率为 1×、2×、 4×。为更好收敛，这里的模型训练1000轮。

线性评估。表6将我们的结果与此前方法比较（Zhuang 等，2019；He 等，2019；Misra 与 van der Maaten，2019；Hénaff 等，2019；Kolesnikov 等，2019；Donahue 与 Simonyan，2019；Bachman 等，

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
Linear evaluation:												
SimCLR (ours)	**76.9**	**95.3**	80.2	48.4	**65.9**	60.0	61.2	**84.2**	**78.9**	89.2	**93.9**	**95.0**
Supervised	75.2	**95.7**	**81.2**	**56.4**	64.9	**68.8**	**63.8**	83.8	**78.7**	**92.3**	**94.1**	94.2
Fine-tuned:												
SimCLR (ours)	**89.4**	**98.6**	**89.0**	**78.2**	**68.1**	**92.1**	**87.0**	**86.6**	**77.8**	92.1	**94.1**	97.6
Supervised	88.7	98.3	**88.7**	**77.8**	67.0	91.4	**88.0**	86.5	**78.8**	**93.2**	**94.2**	**98.0**
Random init	88.3	96.0	81.9	**77.0**	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

Table 8. Comparison of transfer learning performance of our self-supervised approach with supervised baselines across 12 natural image classification datasets, for ResNet-50 ($4\times$) models pretrained on ImageNet. Results not significantly worse than the best ($p > 0.05$, permutation test) are shown in bold. See Appendix B.8 for experimental details and results with standard ResNet-50.

2019; Tian et al., 2019) in the linear evaluation setting (see Appendix B.6). Table 1 shows more numerical comparisons among different methods. We are able to use standard networks to obtain substantially better results compared to previous methods that require specifically designed architectures. The best result obtained with our ResNet-50 ($4\times$) can match the supervised pretrained ResNet-50.

Semi-supervised learning. We follow Zhai et al. (2019) and sample 1% or 10% of the labeled ILSVRC-12 training datasets in a class-balanced way (∼12.8 and ∼128 images per class respectively). [11] We simply fine-tune the whole base network on the labeled data without regularization (see Appendix B.5). Table 7 shows the comparisons of our results against recent methods (Zhai et al., 2019; Xie et al., 2019; Sohn et al., 2020; Wu et al., 2018; Donahue & Simonyan, 2019; Misra & van der Maaten, 2019; Hénaff et al., 2019). The supervised baseline from (Zhai et al., 2019) is strong due to intensive search of hyper-parameters (including augmentation). Again, our approach significantly improves over state-of-the-art with both 1% and 10% of the labels. Interestingly, fine-tuning our pretrained ResNet-50 ($2\times, 4\times$) on *full* ImageNet are also significantly better then training from scratch (up to 2%, see Appendix B.2).

Transfer learning. We evaluate transfer learning performance across 12 natural image datasets in both linear evaluation (fixed feature extractor) and fine-tuning settings. Following Kornblith et al. (2019), we perform hyperparameter tuning for each model-dataset combination and select the best hyperparameters on a validation set. Table 8 shows results with the ResNet-50 ($4\times$) model. When fine-tuned, our self-supervised model significantly outperforms the supervised baseline on 5 datasets, whereas the supervised baseline is superior on only 2 (i.e. Pets and Flowers). On the remaining 5 datasets, the models are statistically tied. Full experimental details as well as results with the standard ResNet-50 architecture are provided in Appendix B.8.

[11] The details of sampling and exact subsets can be found in https://www.tensorflow.org/datasets/catalog/imagenet2012_subset.

7. Related Work

The idea of making representations of an image agree with each other under small transformations dates back to Becker & Hinton (1992). We extend it by leveraging recent advances in data augmentation, network architecture and contrastive loss. A similar consistency idea, but for *class label prediction*, has been explored in other contexts such as semi-supervised learning (Xie et al., 2019; Berthelot et al., 2019).

Handcrafted pretext tasks. The recent renaissance of self-supervised learning began with artificially designed pretext tasks, such as relative patch prediction (Doersch et al., 2015), solving jigsaw puzzles (Noroozi & Favaro, 2016), colorization (Zhang et al., 2016) and rotation prediction (Gidaris et al., 2018; Chen et al., 2019). Although good results can be obtained with bigger networks and longer training (Kolesnikov et al., 2019), these pretext tasks rely on somewhat ad-hoc heuristics, which limits the generality of learned representations.

Contrastive visual representation learning. Dating back to Hadsell et al. (2006), these approaches learn representations by contrasting positive pairs against negative pairs. Along these lines, Dosovitskiy et al. (2014) proposes to treat each instance as a class represented by a feature vector (in a parametric form). Wu et al. (2018) proposes to use a memory bank to store the instance class representation vector, an approach adopted and extended in several recent papers (Zhuang et al., 2019; Tian et al., 2019; He et al., 2019; Misra & van der Maaten, 2019). Other work explores the use of in-batch samples for negative sampling instead of a memory bank (Doersch & Zisserman, 2017; Ye et al., 2019; Ji et al., 2019).

Recent literature has attempted to relate the success of their methods to maximization of mutual information between latent representations (Oord et al., 2018; Hénaff et al., 2019; Hjelm et al., 2018; Bachman et al., 2019). However, it is not clear if the success of contrastive approaches is determined by the mutual information, or by the specific form of the contrastive loss (Tschannen et al., 2019).

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
线性评估：												
SimCLR（本文）	76.9	**95.3**	80.2	48.4	**65.9**	60.0	61.2	**84.2**	78.9	89.2	**93.9**	**95.0**
监督	75.2	**95.7**	**81.2**	**56.4**	64.9	**68.8**	**63.8**	83.8	78.7	92.3	**94.1**	94.2
微调：												
SimCLR（本文）	89.4	**98.6**	**89.0**	**78.2**	**68.1**	**92.1**	**87.0**	**86.6**	77.8	92.1	**94.1**	97.6
监督	88.7	98.3	**88.7**	**77.8**	67.0	91.4	**88.0**	86.5	**78.8**	**93.2**	**94.2**	**98.0**
随机初始化	88.3	96.0	81.9	**77.0**	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

表8：在12个自然图像分类数据集上，将我们的自监督方法与监督基线比较。均为在 ImageNet 上预训练的 ResNet-50（4×）。与最佳结果相比没有显著劣势的结果加粗，采用置换检验 $p > 0.05$。实验细节及标准 ResNet-50 结果见附录B.8。

2019；Tian 等，2019），采用线性评估，详见附录B.6。表1进一步提供数值比较。相比需要专门架构的以往方法，我们使用标准网络就取得明显更好结果。ResNet-50（4×）的最佳结果，与监督预训练的标准 ResNet-50 相当。

半监督学习。我们沿用 Zhai 等（2019）的做法，以类别均衡的方式抽取 ILSVRC-12 有标签训练集的1%或10%，分别相当于每类 ∼12.8 和 ∼128 张图像。[11] 只需在这些有标签数据上微调整个基础网络，不使用正则化，详见附录B.5。表7将结果与近期方法比较（Zhai 等，2019；Xie 等，2019；Sohn 等，2020；Wu 等，2018；Donahue 与 Simonyan，2019；Misra 与 van der Maaten，2019；Hénaff 等，2019）。Zhai 等（2019）的监督基线进行了大量超参数搜索，包括数据增强，因此相当强。使用1%和10%标签时，我们的方法都显著优于此前领先结果。有意思的是，在完整 ImageNet 上微调预训练的 ResNet-50（2×、4×），也明显优于从头训练，最多提升2个百分点，见附录B.2。

迁移学习。我们在12个自然图像数据集上，分别评估固定特征提取器的线性评估与全网络微调。按照 Kornblith 等（2019），为每个模型—数据集组合优化超参数，并在验证集选取最佳设置。表8展示 ResNet-50（4×）结果。微调时，自监督模型在5个数据集上显著优于监督基线，监督基线只在 Pets、Flowers 两个数据集上更好；其余5个在统计上相当。完整实验细节及标准 ResNet-50 结果见附录B.8。

[11] 采样细节及具体子集见 https://www.tensorflow.org/datasets/catalog/imagenet2012_subset。

7 相关工作

使图像在小幅变换下产生一致表征的思想，可追溯至 Becker 与 Hinton（1992）。我们结合近期的数据增强、架构与对比损失进展，扩展这一思想。类似的一致性思想，也在半监督等设置中用于类别标签预测（Xie 等，2019；Berthelot 等，2019）。

手工设计的代理任务。近期自监督学习的复兴，始于人工设计的代理任务，包括相对图像块位置预测（Doersch 等，2015）、拼图（Noroozi 与 Favaro，2016）、着色（Zhang 等，2016）和旋转预测（Gidaris 等，2018；Chen 等，2019）。虽然扩大网络和延长训练可取得好结果（Kolesnikov 等，2019），这些任务仍依赖带有一定随意性的启发式规则，限制了表征的通用性。

对比式视觉表征学习。这类方法可追溯至 Hadsell 等（2006），通过比较正样本对与负样本对学习表征。Dosovitskiy 等（2014）提出将每个实例视为一个类别，并以参数化特征向量表示。Wu 等（2018）使用记忆库保存实例类别表征，近期多项工作采用并扩展了这一做法（Zhuang 等，2019；Tian 等，2019；He 等，2019；Misra 与 van der Maaten，2019）。其他研究则使用批内样本进行负采样，而非记忆库（Doersch 与 Zisserman，2017；Ye 等，2019；Ji 等，2019）。

近期文献尝试将方法的成功归因于最大化潜在表征之间的互信息（Oord 等，2018；Hénaff 等，2019；Hjelm 等，2018；Bachman 等，2019）。然而，对比方法的成功究竟由互信息决定，还是由对比损失的具体形式决定，目前并不明确（Tschannen 等，2019）。

We note that almost all individual components of our framework have appeared in previous work, although the specific instantiations may be different. The superiority of our framework relative to previous work is not explained by any single design choice, but by their composition. We provide a comprehensive comparison of our design choices with those of previous work in Appendix C.

8. Conclusion

In this work, we present a simple framework and its instantiation for contrastive visual representation learning. We carefully study its components, and show the effects of different design choices. By combining our findings, we improve considerably over previous methods for self-supervised, semi-supervised, and transfer learning.

Our approach differs from standard supervised learning on ImageNet only in the choice of data augmentation, the use of a nonlinear head at the end of the network, and the loss function. The strength of this simple framework suggests that, despite a recent surge in interest, self-supervised learning remains undervalued.

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我们注意到，框架几乎所有单独组件都曾出现在以往工作中，尽管具体实现可能不同。相较以往方法的优势，并不能由某个单独选择解释，而来自这些选择的组合。附录C全面比较了我们与此前工作的设计。

8 结论

本文提出用于视觉表征对比学习的简单框架及其具体实现。我们仔细研究各组件，展示不同设计选择的影响。结合这些发现，在自监督、半监督及迁移学习中，相较此前方法取得明显改善。

与标准 ImageNet 监督学习相比，我们只改变了数据增强、网络末端的非线性投影头，以及损失函数。这个简单框架取得的强表现说明：尽管近期关注快速增长，自监督学习仍被低估。

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Zhuang, C., Zhai, A. L., and Yamins, D. Local aggregation for unsupervised learning of visual embeddings. In *Proceedings of the IEEE International Conference on Computer Vision*, pp. 6002–6012, 2019.

Wu, Z., Xiong, Y., Yu, S. X., and Lin, D. Unsupervised feature learning via non-parametric instance discrimination. In Proceed- ings of the IEEE Conference o n Computer Vision and Pattern Recognition, pp. 3733–3742, 2018. Xiao, J., Hays, J., Ehinger, K. A., Oliva, A., and Torralba, A. Sun database: Large-scale scene recognition from abbey to zoo. In IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3485–3492. IEEE, 2010. Xie, Q., Dai, Z., Hovy, E., Luong, M.-T., and Le, Q. V. Unsu- pervised data augmentation. arXiv preprint arXiv:1904 12848, 2019 . . Ye, M., Zhang, X., Yuen, P. C., and Chang, S.-F. Unsupervised embedding learning via invariant and spreading instance feature. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 6210–6219, 2019. You, Y., Gitman, I., and Ginsburg, B. Large batch training of con- volutional networks. arXiv preprint arXiv:1708.03888, 2017. Zhai, X., Oliver, A., Kolesnikov, A., and Beyer, L. S4l: Self-supervised semi-supervised learning. In The IEEE International Conference on Computer Vision (ICCV), October 2019. Zhang, R., Isola, P., and Efros, A. A. Colorful image coloriza- tion. In European conference on co mputer vision, pp. 649–666. Springer, 2016. Zhuang, C., Zhai, A. L., and Yamins, D. Local aggregation for unsupervised learning of visual embeddings. In Proceedings of the IEEE International Conference on Computer Vision, pp. 6002–6012, 2019.

A. Data Augmentation Details

In our default pretraining setting (which is used to train our best models), we utilize random crop (with resize and random flip), random color distortion, and random Gaussian blur as the data augmentations. The details of these three augmentations are provided below.

Random crop and resize to 224x224 We use standard Inception-style random cropping (Szegedy et al., 2015). The crop of random size (uniform from 0.08 to 1.0 in area) of the original size and a random aspect ratio (default: of 3/4 to 4/3) of the original aspect ratio is made. This crop is finally resized to the original size. This has been implemented in Tensorflow as "slim.preprocessing.inception_preprocessing.distorted_bounding_box_crop", or in Pytorch as "torchvision.transforms.RandomResizedCrop". Additionally, the random crop (with resize) is always followed by a random horizontal/left-to-right flip with 50% probability. This is helpful but not essential. By removing this from our default augmentation policy, the top-1 linear evaluation drops from 64.5% to 63.4% for our ResNet-50 model trained in 100 epochs.

Color distortion Color distortion is composed by color jittering and color dropping. We find stronger color jittering usually helps, so we set a strength parameter.

A pseudo-code for color distortion using TensorFlow is as follows.

```
import tensorflow as tf
def color_distortion(image, s=1.0):
    # image is a tensor with value range in [0, 1].
    # s is the strength of color distortion.

    def color_jitter(x):
        # one can also shuffle the order of following augmentations
        # each time they are applied.
        x = tf.image.random_brightness(x, max_delta=0.8*s)
        x = tf.image.random_contrast(x, lower=1-0.8*s, upper=1+0.8*s)
        x = tf.image.random_saturation(x, lower=1-0.8*s, upper=1+0.8*s)
        x = tf.image.random_hue(x, max_delta=0.2*s)
        x = tf.clip_by_value(x, 0, 1)
        return x

    def color_drop(x):
        image = tf.image.rgb_to_grayscale(image)
        image = tf.tile(image, [1, 1, 3])

    # randomly apply transformation with probability p.
    image = random_apply(color_jitter, image, p=0.8)
    image = random_apply(color_drop, image, p=0.2)
    return image
```

A pseudo-code for color distortion using Pytorch is as follows [12].

```
from torchvision import transforms
def get_color_distortion(s=1.0):
    # s is the strength of color distortion.
    color_jitter = transforms.ColorJitter(0.8*s, 0.8*s, 0.8*s, 0.2*s)
    rnd_color_jitter = transforms.RandomApply([color_jitter], p=0.8)
    rnd_gray = transforms.RandomGrayscale(p=0.2)
    color_distort = transforms.Compose([
        rnd_color_jitter,
        rnd_gray])
```

[12] Our code and results are based on Tensorflow, the Pytorch code here is a reference.

A 数据增强细节

默认预训练设置用于训练最佳模型，其增强包含随机裁剪、缩放与翻转、随机颜色扰动，以及随机高斯模糊。三种增强详述如下。

随机裁剪并缩放至224×224。我们采用标准的 Inception 风格随机裁剪（Szegedy 等，2015），裁剪面积为原图面积的0.08至1.0倍，均匀采样；宽高比随机选择，默认范围为3/4至4/3。最后将裁剪图像缩放至原尺寸。TensorFlow 实现为"slim.preprocessing.inception_preprocessing.distorted_bounding_box_crop"，PyTorch 实现为"torchvision.transforms.RandomResizedCrop"。裁剪缩放后，还以50%概率水平翻转。翻转有帮助，但并非必需。移除默认策略中的翻转后，训练100轮的 ResNet-50 的 top-1 线性评估准确率从64.5%降至63.4%。

颜色扰动。它由颜色抖动和去色组成。更强的颜色抖动通常有帮助，因此我们设置了强度参数。

使用 TensorFlow 实现颜色扰动的伪代码如下。

import tensorflow as tf def color_distortion(image, s=1.0): # image isa tensor with value range in [0, 1]. # s is thestrength of color distortion.

def color_jitter(x): # one can also shuffle the order of following augmentations # each time theyareapplied. x = tf.image.random_brightness(x, max_delta=0.8*s) x = tf.image.random_contrast(x, lower=1-0.8*s, upper=1+0.8*s) x = tf.image.random_saturation(x, lower=1-0.8*s, upper=1+0.8*s) x = tf.image.random_hue(x, max_delta=0.2*s) x = tf.clip_by_value(x, 0, 1) return x

def color_drop(x): image = tf.image.rgb_to_grayscale(image) image = tf.tile(image, [1, 1, 3])

randomly apply transformation with probability p. image = random_apply(color_jitter, image, p=0.8) image = rando m_apply(color_drop, image, p=0.2) return image

使用 PyTorch 实现颜色扰动的伪代码如下 [12]。

from torchvision import transforms def get_color_distortion(s=1.0): # s is the strength of color distortion. color_jitter =transforms.ColorJitter(0.8*s, 0.8*s, 0.8*s, 0.2*s) rnd_color_ jitter = transforms.RandomApply([color_jitter], p=0.8) rnd_gray = transforms.RandomGrayscale(p=0.2) color_distort =transforms.Compose([rnd_color_ jitter, rnd_gray])[12]我们的代码与结果基于 TensorFlow，这里的 PyTorch 代码仅供参考。

```
return color_distort
```

Gaussian blur This augmentation is in our default policy. We find it helpful, as it improves our ResNet-50 trained for 100 epochs from 63.2% to 64.5%. We blur the image 50% of the time using a Gaussian kernel. We randomly sample $\sigma \in [0.1, 2.0]$, and the kernel size is set to be 10% of the image height/width.

B. Additional Experimental Results

B.1. Batch Size and Training Steps

Figure B.1 shows the top-5 accuracy on linear evaluation when trained with different batch sizes and training epochs. The conclusion is very similar to top-1 accuracy shown before, except that the differences between different batch sizes and training steps seems slightly smaller here.

In both Figure 9 and Figure B.1, we use a linear scaling of learning rate similar to (Goyal et al., 2017) when training with different batch sizes. Although linear learning rate scaling is popular with SGD/Momentum optimizer, we find a square root learning rate scaling is more desirable with LARS optimizer. With square root learning rate scaling, we have $\text{LearningRate} = 0.075 \times \sqrt{\text{BatchSize}}$, instead of $\text{LearningRate} = 0.3 \times \text{BatchSize}/256$ in the linear scaling case, but the learning rate is the same under both scaling methods when batch size of 4096 (our default batch size). A comparison is presented in Table B.1, where we observe that square root learning rate scaling improves the performance for models trained with small batch sizes and in smaller number of epochs.

Batch size \ Epochs	100	200	400	800
256	57.5 / **62.8**	61.9 / **64.3**	64.7 / **65.7**	66.6 / 66.5
512	60.7 / **63.8**	64.0 / **65.6**	66.2 / 66.7	67.8 / 67.4
1024	62.8 / **64.3**	65.3 / **66.1**	67.2 / 67.2	68.5 / 68.3
2048	64.0 / **64.7**	66.1 / **66.8**	68.1 / 67.9	68.9 / 68.8
4096	64.6 / 64.5	66.5 / 66.8	68.2 / 68.0	68.9 / 69.1
8192	64.8 / 64.8	66.6 / 67.0	67.8 / 68.3	69.0 / 69.1

Table B.1. Linear evaluation (top-1) under different batch sizes and training epochs. On the left side of slash sign are models trained with linear LR scaling, and on the right are models trained with square root LR scaling. The result is bolded if it is more than 0.5% better. Square root LR scaling works better for smaller batch size trained in fewer epochs (with LARS optimizer).

We also train with larger batch size (up to 32K) and longer (up to 3200 epochs), with the square root learning rate scaling. A shown in Figure B.2, *the performance seems to saturate with a batch size of 8192, while training longer can still significantly improve the performance*.

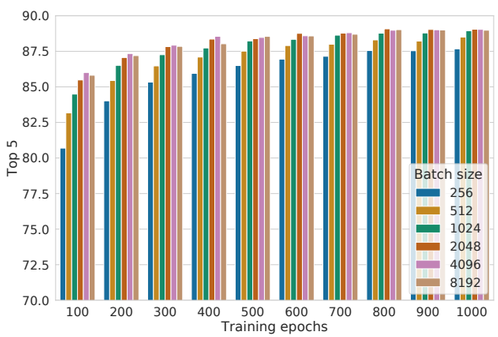


Figure B.1. Linear evaluation (top-5) of ResNet-50 trained with different batch sizes and epochs. Each bar is a single run from scratch. See Figure 9 for top-1 accuracy.

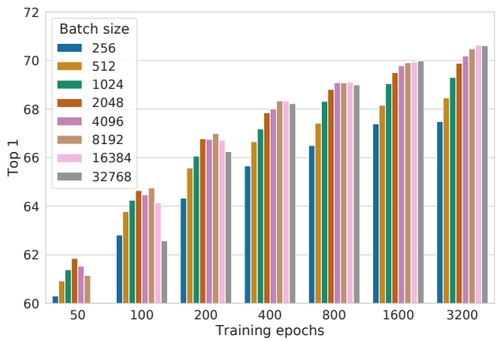


Figure B.2. Linear evaluation (top-1) of ResNet-50 trained with different batch sizes and *longer* epochs. Here a *square root* learning rate, instead of a linear one, is utilized.

return color distort_

高斯模糊。这是默认增强之一，可将训练100轮的 ResNet-50 的准确率从63.2%提高至64.5%。我们以50%概率使用高斯核模糊图像，随机采样 $\sigma \in [0.1, 2.0]$，核尺寸设为图像高度或宽度的10%。

B 补充实验结果

B.1 批量大小与训练步数

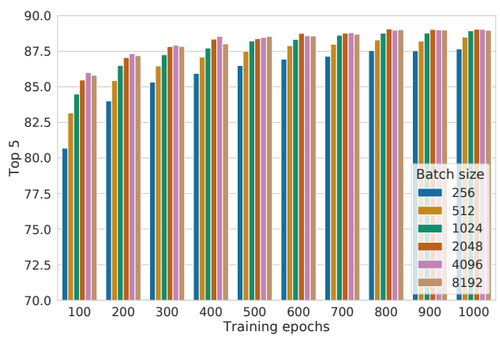
图B.1展示不同批量、训练轮数下的线性评估 top-5。结论与前述 top-1 相近，但不同批量和训练步数之间的差异略小。

图9和图B.1中，不同批量大小采用与 Goyal 等（2017）类似的线性学习率缩放。尽管这一方式常用于 SGD/动量优化器，我们发现 LARS 更适合平方根缩放。此时 $\text{LearningRate} = 0.075 \times \sqrt{\text{BatchSize}}$，而非线性缩放的 $\text{LearningRate} = 0.3 \times \text{B}\sout{\text{atchSize}}/256$ ；在默认批量 4096 时，两种方式学习率相同。表 B.1 比较发现，平方根缩放提升了小批量、较少训练轮数下的性能。

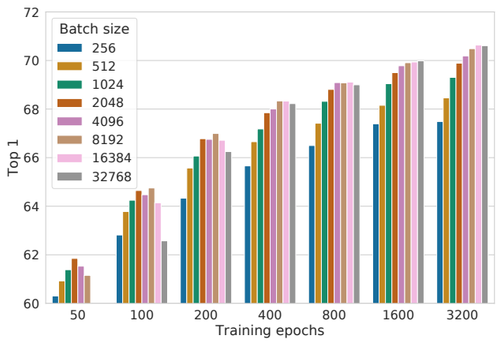
批量大小 / 轮数	100	200	400	800
256	57.5 / 62.8	61.9 / 64.3	64.7 / 65.7	66.6 / 66.5
512	60.7 / 63.8	64.0 / 65.6	66.2 / 66.7	67.8 / 67.4
1024	62.8 / 64.3	65.3 / 66.1	67.2 / 67.2	68.5 / 68.3
2048	64.0 / 64.7	66.1 / 66.8	68.1 / 67.9	68.9 / 68.8
4096	64.6 / 64.5	66.5 / 66.8	68.2 / 68.0	68.9 / 69.1
8192	64.8 / 64.8	66.6 / 67.0	67.8 / 68.3	69.0 / 69.1

表B.1：不同批量和训练轮数下的线性评估 top-1 准确率。斜杠左侧采用线性学习率缩放，右侧采用平方根缩放。高出0.5个百分点以上的结果加粗。在 LARS 优化器下，小批量且训练较少轮时，平方根缩放更好。

我们还采用平方根学习率缩放，将批量增至最多32K，训练延长至最多3200轮。图B.2表明，批量8192时性能似乎趋于饱和，但延长训练仍能明显
改善性能。



图B.1：不同批量与训练轮数下，ResNet-50 的线性评估 top-5。每根柱表示一次从头训练，top-1 见图9。



图B.2：不同批量和更长训练下，ResNet-50 的线性评估 top-1。这里使用平方根而非线性学习率缩放。

B.2. Broader composition of data augmentations further improves performance

Our best results in the main text (Table 6 and 7) can be further improved when expanding the default augmentation policy to include the following: (1) Sobel filtering, (2) additional color distortion (equalize, solarize), and (3) motion blur. For linear evaluation protocol, the ResNet-50 models ($1\times, 2\times, 4\times$) trained with broader data augmentations achieve 70.0 (+0.7), 74.4 (+0.2), 76.8 (+0.3), respectively.

Table B.2 shows ImageNet accuracy obtained by fine-tuning the SimCLR model (see Appendix B.5 for the details of fine-tuning procedure). Interestingly, when fine-tuned on full (100%) ImageNet training set, our ResNet ($4\times$) model achieves 80.4% top-1 / 95.4% top-5 [13], which is significantly better than that (78.4% top-1 / 94.2% top-5) of training from scratch using the same set of augmentations (i.e. random crop and horizontal flip). For ResNet-50 ($2\times$), fine-tuning our pre-trained ResNet-50 ($2\times$) is also better than training from scratch (77.8% top-1 / 93.9% top-5). There is no improvement from fine-tuning for ResNet-50.

Architecture	Label fraction					
	1%		10%		100%	
	Top 1	Top 5	Top 1	Top 5	Top 1	Top 5
ResNet-50	49.4	76.6	66.1	88.1	76.0	93.1
ResNet-50 ($2\times$)	59.4	83.7	71.8	91.2	79.1	94.8
ResNet-50 ($4\times$)	64.1	86.6	74.8	92.8	80.4	95.4

Table B.2. Classification accuracy obtained by fine-tuning the SimCLR (which is pretrained with broader data augmentations) on 1%, 10% and full of ImageNet. As a reference, our ResNet-50 ($4\times$) trained from scratch on 100% labels achieves 78.4% top-1 / 94.2% top-5.

B.3. Effects of Longer Training for Supervised Models

Here we perform experiments to see how training steps and stronger data augmentation affect supervised training. We test ResNet-50 and ResNet-50 ($4\times$) under the same set of data augmentations (random crops, color distortion, 50% Gaussian blur) as used in our unsupervised models. Figure B.3 shows the top-1 accuracy. We observe that there is no significant benefit from training supervised models longer on ImageNet. Stronger data augmentation slightly improves the accuracy of ResNet-50 ($4\times$) but does not help on ResNet-50. When stronger data augmentation is applied, ResNet-50 generally requires longer training (e.g. 500 epochs [14]) to obtain the optimal result, while ResNet-50 ($4\times$) does not benefit from longer training.

Model	Training epochs	Top 1		
		Crop	+Color	+Color+Blur
ResNet-50	90	76.5	75.6	75.3
	500	76.2	76.5	76.7
	1000	75.8	75.2	76.4
ResNet-50 ($4\times$)	90	78.4	78.9	78.7
	500	78.3	78.4	78.5
	1000	77.9	78.2	78.3

Table B.3. Top-1 accuracy of supervised models trained longer under various data augmentation procedures (from the same set of data augmentations for contrastive learning).

B.4. Understanding The Non-Linear Projection Head

Figure B.3 shows the eigenvalue distribution of linear projection matrix $W \in R^{2048\times2048}$ used to compute $\boldsymbol{z} = W\boldsymbol{h}$. This matrix has relatively few large eigenvalues, indicating that it is approximately low-rank.

Figure B.4 shows t-SNE (Maaten & Hinton, 2008) visualizations of $\boldsymbol{h}$ and $\boldsymbol{z} = g(\boldsymbol{h})$ for randomly selected 10 classes by our best ResNet-50 (top-1 linear evaluation 69.3%). Classes represented by $\boldsymbol{h}$ are better separated compared to $\boldsymbol{z}$.

[13] It is 80.1% top-1 / 95.2% top-5 without broader augmentations for pretraining SimCLR.
[14] With AutoAugment (Cubuk et al., 2019), optimal test accuracy can be achieved between 900 and 500 epochs.

B.2 更广泛的增强组合进一步改善性能

在默认增强策略中进一步加入（1）Sobel 滤波、（2）额外颜色扰动，如直方图均衡和过曝反相、（3）运动模糊，可进一步提升正文表6、7中的最佳结果。采用更广泛增强的 ResNet-50（1×、2×、4×），线性评估分别达到70.0（+0.7）、74.4（+0.2）、76.8（+0.3）。

表B.2展示微调 SimCLR 后的 ImageNet 准确率，微调细节见附录B.5。有意思的是，在完整训练集上微调，ResNet（4×）达到80.4%的 top-1 与95.4%的 top-5 [13],，明显优于使用相同增强，即随机裁剪和水平翻转，从头训练所得的78.4%与94.2%。对于 ResNet-50（2×），微调预训练 ResNet-50（2×）也优于从头训练所得的77.8% top-1 和93.9% top-5。但标准 ResNet-50 的微调没有带来改进。

架构	标签比例					
	1%		10%		100%	
	Top 1	Top 5	Top 1	Top 5	Top 1	Top 5
ResNet-50	49.4	76.6	66.1	88.1	76.0	93.1
ResNet-50 (2×)	59.4	83.7	71.8	91.2	79.1	94.8
ResNet-50 (4×)	64.1	86.6	74.8	92.8	80.4	95.4

表B.2：在1%、10%及全部 ImageNet 标签上，微调采用更广泛增强预训练的 SimCLR，所得分类准确率。作为参考，ResNet-50（4×）用全部标签从头训练，达到78.4% top-1、94.2% top-5。

B.3 更长训练对监督模型的影响

这里通过实验研究训练步数与更强增强对监督训练的影响。ResNet-50 与 ResNet-50（4×）采用与无监督模型相同的增强：随机裁剪、颜色扰动、50%概率的高斯模糊。图B.3展示 top-1 准确率。可以看到，在 ImageNet 上延长监督训练没有显著收益。更强增强略微提高 ResNet-50（4×）准确率，但无助于标准 ResNet-50。采用更强增强时，标准 ResNet-50 通常需要更长训练，如500轮 [14]，才能取得最佳结果，而 ResNet-50（4×）未受益于更长训练。

模型	训练轮数	Top 1		
		Crop	+颜色	+颜色+模糊
ResNet-50	90	76.5	75.6	75.3
	500	76.2	76.5	76.7
	1000	75.8	75.2	76.4
ResNet-50 (4×)	90	78.4	78.9	78.7
	500	78.3	78.4	78.5
	1000	77.9	78.2	78.3

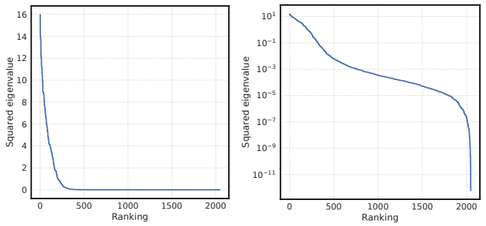
表B.3：使用不同增强策略、训练更久的监督模型的 top-1。增强来自与对比学习相同的操作集合。

B.4 理解非线性投影头

图B.3展示用于计算 $\boldsymbol{z} = W\boldsymbol{h}$ 的线性投影矩阵 $W \in R^{2048\times2048}$ 的特征值分布。较大的特征值很少，说明该矩阵近似低秩。

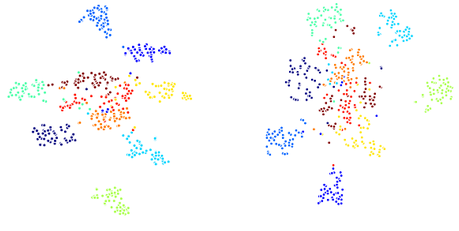
图B.4展示最佳标准 ResNet-50 的 $\boldsymbol{h}$ 与 $\boldsymbol{z} = g(\boldsymbol{h})$ 在随机选取10类上的 t-SNE 可视化（Maaten 与 Hinton, 2008）。该模型 top-1 线性评估为69.3%。相比 $\boldsymbol{z}$，$\boldsymbol{h}$ 表示的类别分离得更好。

[13] 若 SimCLR 预训练不使用更广泛增强，则为80.1% top-1、95.2% top-5。[14] 使用 AutoAugment（Cubuk 等，2019）时，在500至900轮之间可达到最佳测试准确率。



(a) Y-axis in uniform scale. (b) Y-axis in log scale.

Figure B.3. Squared real eigenvalue distribution of linear projection matrix $W \in R^{2048\times2048}$ used to compute $g(\boldsymbol{h}) = W\boldsymbol{h}$.



(a) $\boldsymbol{h}$ (b) $\boldsymbol{z} = g(\boldsymbol{h})$

Figure B.4. t-SNE visualizations of hidden vectors of images from a randomly selected 10 classes in the validation set.

B.5. Semi-supervised Learning via Fine-Tuning

Fine-tuning Procedure We fine-tune using the Nesterov momentum optimizer with a batch size of 4096, momentum of 0.9, and a learning rate of 0.8 (following $\mathrm{LearningRate} = 0.05 \times \mathrm{BatchSize}/256$) without warmup. Only random cropping (with random left-to-right flipping and resizing to 224x224) is used for preprocessing. We do not use any regularization (including weight decay). For 1% labeled data we fine-tune for 60 epochs, and for 10% labeled data we fine-tune for 30 epochs. For the inference, we resize the given image to 256x256, and take a single center crop of 224x224.

Table B.4 shows the comparisons of top-1 accuracy for different methods for semi-supervised learning. Our models significantly improve state-of-the-art.

Method	Architecture	Label fraction (Top 1) 1%	10%
Supervised baseline	ResNet-50	25.4	56.4
Methods using label-propagation:			
UDA (w. RandAug)	ResNet-50	-	68.8
FixMatch (w. RandAug)	ResNet-50	-	71.5
S4L (Rot+VAT+Ent. Min.)	ResNet-50 (4×)	-	73.2
Methods using self-supervised representation learning only:			
CPC v2	ResNet-161(∗)	52.7	73.1
SimCLR (ours)	ResNet-50	48.3	65.6
SimCLR (ours)	ResNet-50 (2×)	58.5	71.7
SimCLR (ours)	ResNet-50 (4×)	**63.0**	**74.4**

Table B.4. ImageNet top-1 accuracy of models trained with few labels. See Table 7 for top-5 accuracy.

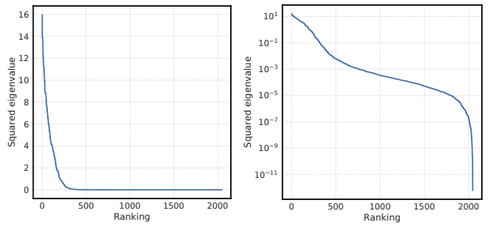
B.6. Linear Evaluation

For linear evaluation, we follow similar procedure as fine-tuning (described in Appendix B.5), except that a larger learning rate of 1.6 (following $\mathrm{LearningRate} = 0.1 \times \mathrm{BatchSize}/256$) and longer training of 90 epochs. Alternatively, using LARS optimizer with the pretraining hyper-parameters also yield similar results. Furthermore, we find that attaching the linear classifier on top of the base encoder (with a $\mathrm{stop_gradient}$ on the input to linear classifier to prevent the label information from influencing the encoder) and train them simultaneously during the pretraining achieves similar performance.

B.7. Correlation Between Linear Evaluation and Fine-Tuning

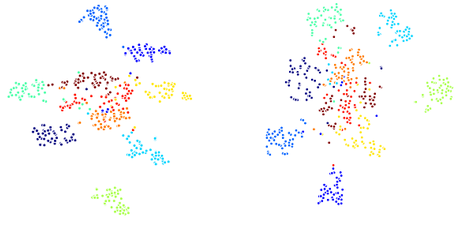
Here we study the correlation between linear evaluation and fine-tuning under different settings of training step and network architecture.

Figure B.5 shows linear evaluation versus fine-tuning when training epochs of a ResNet-50 (using batch size of 4096) are varied from 50 to 3200 as in Figure B.2. While they are almost linearly correlated, it seems fine-tuning on a small fraction



(a) Y-axis in uniform scale. (b) Y-axis in log scale.

图B.3：计算 $g(\boldsymbol{h}) = W\boldsymbol{h}$ 所用线性投影矩阵 $W \in R^{2048\times2048}$ 的实特征值平方分布。



(a) $\boldsymbol{h}$ (b) $\boldsymbol{z} = g(\boldsymbol{h})$

图B.4：验证集中随机选取10类图像，其隐藏向量的 t-SNE 可视化。

B.5 通过微调进行半监督学习

微调流程。使用 Nesterov 动量优化器，批量4096、动量0.9、学习率0.8，遵循 $\mathrm{LearningRate} = 0.05 \times \mathrm{BatchSize}/256$，不预热。预处理仅用随机裁剪、随机水平翻转和缩放到224×224。不使用任何正则化，包括权重衰减。1%标签微调60轮，10%标签微调30轮。推理时先将图像缩放到256×256，再取224×224的单个中心裁剪。

表B.4比较不同半监督方法的 top-1。我们的模型显著改善此前最佳结果。

方法	架构	标签比例 (Top 1) 1%	10%
监督基线	ResNet-50	25.4	56.4
使用标签传播的方法：			
UDA (w. RandAug)	ResNet-50	-	68.8
FixMatch (w. RandAug)	ResNet-50	-	71.5
S4L (Rot+VAT+Ent. Min.)	ResNet-50 (4×)	-	73.2
只使用自监督表征学习的方法：			
CPC v2	ResNet-161(∗)	52.7	73.1
SimCLR（本文）	ResNet-50	48.3	65.6
SimCLR（本文）	ResNet-50 (2×)	58.5	71.7
SimCLR（本文）	ResNet-50 (4×)	**63.0**	**74.4**

表B.4：少量标签训练模型的 ImageNet top-1，top-5 见表7。

B.6 线性评估

线性评估采用与微调相似的流程，见附录B.5，但使用更大的1.6学习率，遵循 $\mathrm{LearningRate} = 0.1 \times \mathrm{BatchSize}/256$，并训练90轮。用 LARS 及预训练超参数也能得到相近结果。此外，将线性分类器接到基础编码器上，在分类器输入处使用 $\mathrm{stop_gradient}$，阻止标签信息影响编码器，并在预训练时同步训练两者，也能取得相近表现。

B.7 线性评估与微调的相关性

这里研究不同训练步数与网络架构下，线性评估与微调之间的相关性。

图B.5比较线性评估与微调：如图B.2，ResNet-50 以4096批量训练50至3200轮。虽然两者几乎线性相关，但似乎在少量

of labels benefits more from training longer.

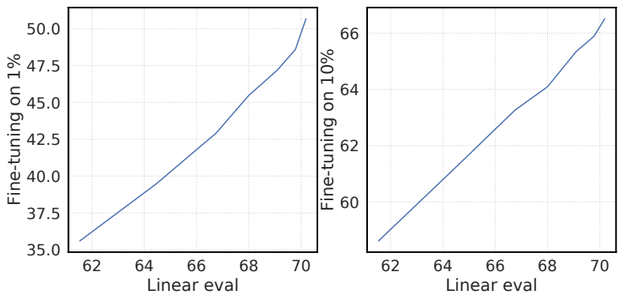


Figure B.5. Top-1 accuracy of models trained in different epochs (from Figure B.2), under linear evaluation and fine-tuning.

Figure B.6 shows shows linear evaluation versus fine-tuning for different architectures of choice.

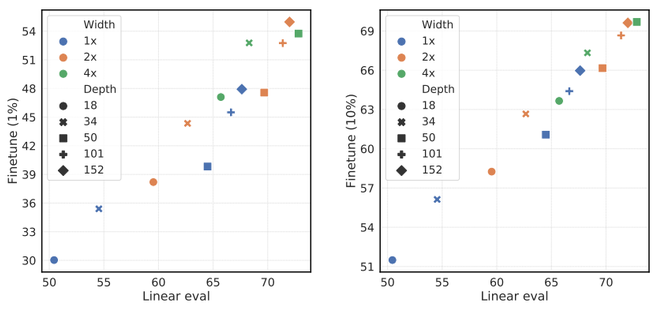


Figure B.6. Top-1 accuracy of different architectures under linear evaluation and fine-tuning.

B.8. Transfer Learning

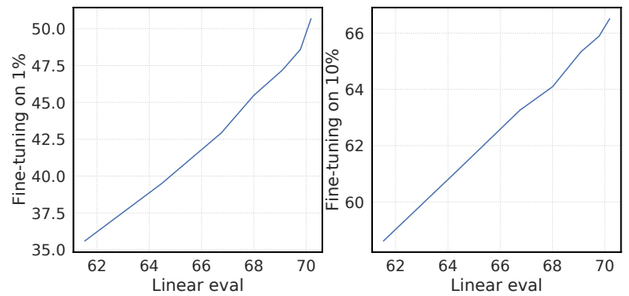
We evaluated the performance of our self-supervised representation for transfer learning in two settings: linear evaluation, where a logistic regression classifier is trained to classify a new dataset based on the self-supervised representation learned on ImageNet, and fine-tuning, where we allow all weights to vary during training. In both cases, we follow the approach described by Kornblith et al. (2019), although our preprocessing differs slightly.

B.8.1. Methods

Datasets We investigated transfer learning performance on the Food-101 dataset (Bossard et al., 2014), CIFAR-10 and CIFAR-100 (Krizhevsky & Hinton, 2009), Birdsnap (Berg et al., 2014), the SUN397 scene dataset (Xiao et al., 2010), Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), the PASCAL VOC 2007 classification task (Everingham et al., 2010), the Describable Textures Dataset (DTD) (Cimpoi et al., 2014), Oxford-IIIT Pets (Parkhi et al., 2012), Caltech-101 (Fei-Fei et al., 2004), and Oxford 102 Flowers (Nilsback & Zisserman, 2008). We follow the evaluation protocols in the papers introducing these datasets, *i.e.*, we report top-1 accuracy for Food-101, CIFAR-10, CIFAR-100, Birdsnap, SUN397, Stanford Cars, and DTD; mean per-class accuracy for FGVC Aircraft, Oxford-IIIT Pets, Caltech-101, and Oxford 102 Flowers; and the 11-point mAP metric as defined in Everingham et al. (2010) for PASCAL VOC 2007. For DTD and SUN397, the dataset creators defined multiple train/test splits; we report results only for the first split. Caltech-101 defines no train/test split, so we randomly chose 30 images per class and test on the remainder, for fair comparison with previous work (Donahue et al., 2014; Simonyan & Zisserman, 2014).

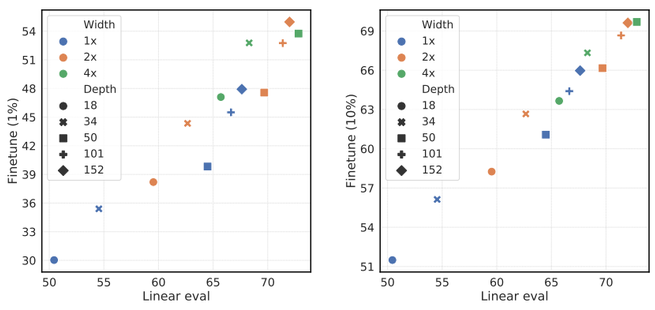
We used the validation sets specified by the dataset creators to select hyperparameters for FGVC Aircraft, PASCAL VOC

标签上微调，更能受益于延长训练。



图B.5：模型的 top-1，　　不同训练轮数的模型，来自图B.2，采用　　线性评估与微调。

Fi图B.6展示不同架构的线性评估与微调，　　.



图B.6：不同架构的线性评估及微调 top-1。

B.8 迁移学习

我们在两种设置中评估自监督表征的迁移性能：线性评估，在 ImageNet 自监督表征上训练逻辑回归分类器，分类新数据集；以及微调，允许训练时更新全部权重。两者均遵循 Kornblith 等（2019）的方法，但预处理略有不同。

B.8.1 方法

数据集。我们在 Food-101（Bossard 等，2014）、CIFAR-10/100（Krizhevsky 与 Hinton，2009）、Birdsnap（Berg 等，2014）、SUN397 场景数据集（Xiao 等，2010）、Stanford Cars（Krause 等，2013）、FGVC Aircraft（Maji 等，2013）、PASCAL VOC 2007 分类任务（Everingham 等，2010）、可描述纹理数据集 DTD（Cimpoi 等，2014）、Oxford-IIIT Pets（Parkhi 等，2012）、Caltech-101（Fei-Fei 等，2004）及 Oxford 102 Flowers（Nilsback 与 Zisserman，2008）上研究迁移学习。评估遵循各数据集原论文：Food-101、CIFAR-10/100、Birdsnap、SUN397、Stanford Cars、DTD 报告 top-1 准确率；FGVC Aircraft、Oxford-IIIT Pets、Caltech-101、Oxford 102 Flowers 报告类别平均准确率；PASCAL VOC 2007 使用 Everingham 等（2010）定义的11点 mAP。DTD 和 SUN397 提供多种训练/测试划分，我们只报告第一种。Caltech-101 没有固定划分，因此每类随机选30张图像训练，其余测试，以便与此前工作公平比较（Donahue 等，2014；Simonyan 与 Zisserman，2014）。

我们使用验证集　　按照数据集创建者提供的划分，为 FGVC Aircraft、PASCAL VOC　　C

2007, DTD, and Oxford 102 Flowers. For other datasets, we held out a subset of the training set for validation while performing hyperparameter tuning. After selecting the optimal hyperparameters on the validation set, we retrained the model using the selected parameters using all training and validation images. We report accuracy on the test set.

Transfer Learning via a Linear Classifier We trained an ℓ_2-regularized multinomial logistic regression classifier on features extracted from the frozen pretrained network. We used L-BFGS to optimize the softmax cross-entropy objective and we did not apply data augmentation. As preprocessing, all images were resized to 224 pixels along the shorter side using bicubic resampling, after which we took a 224×224 center crop. We selected the ℓ_2 regularization parameter from a range of 45 logarithmically spaced values between 10^{-6} and 10^{5}.

Transfer Learning via Fine-Tuning We fine-tuned the entire network using the weights of the pretrained network as initialization. We trained for 20,000 steps at a batch size of 256 using SGD with Nesterov momentum with a momentum parameter of 0.9. We set the momentum parameter for the batch normalization statistics to $\max(1 - 10/s, 0.9)$ where s is the number of steps per epoch. As data augmentation during fine-tuning, we performed only random crops with resize and flips; in contrast to pretraining, we did not perform color augmentation or blurring. At test time, we resized images to 256 pixels along the shorter side and took a 224×224 center crop. (Additional accuracy improvements may be possible with further optimization of data augmentation, particularly on the CIFAR-10 and CIFAR-100 datasets.) We selected the learning rate and weight decay, with a grid of 7 logarithmically spaced learning rates between 0.0001 and 0.1 and 7 logarithmically spaced values of weight decay between 10^{-6} and 10^{-3}, as well as no weight decay. We divide these values of weight decay by the learning rate.

Training from Random Initialization We trained the network from random initialization using the same procedure as for fine-tuning, but for longer, and with an altered hyperparameter grid. We chose hyperparameters from a grid of 7 logarithmically spaced learning rates between 0.001 and 1.0 and 8 logarithmically spaced values of weight decay between 10^{-5} and $10^{-1.5}$. Importantly, our random initialization baselines are trained for 40,000 steps, which is sufficiently long to achieve near-maximal accuracy, as demonstrated in Figure 8 of Kornblith et al. (2019).

On Birdsnap, there are no statistically significant differences among methods, and on Food-101, Stanford Cars, and FGVC Aircraft datasets, fine-tuning provides only a small advantage over training from random initialization. However, on the remaining 8 datasets, pretraining has clear advantages.

Supervised Baselines We compare against architecturally identical ResNet models trained on ImageNet with standard cross-entropy loss. These models are trained with the same data augmentation as our self-supervised models (crops, strong color augmentation, and blur) and are also trained for 1000 epochs. We found that, although stronger data augmentation and longer training time do not benefit accuracy on ImageNet, these models performed significantly better than a supervised baseline trained for 90 epochs and ordinary data augmentation for linear evaluation on a subset of transfer datasets. The supervised ResNet-50 baseline achieves 76.3% top-1 accuracy on ImageNet, vs. 69.3% for the self-supervised counterpart, while the ResNet-50 ($4\times$) baseline achieves 78.3%, vs. 76.5% for the self-supervised model.

Statistical Significance Testing We test for the significance of differences between model with a permutation test. Given predictions of two models, we generate 100,000 samples from the null distribution by randomly exchanging predictions for each example and computing the difference in accuracy after performing this randomization. We then compute the percentage of samples from the null distribution that are more extreme than the observed difference in predictions. For top-1 accuracy, this procedure yields the same result as the exact McNemar test. The assumption of exchangeability under the null hypothesis is also valid for mean per-class accuracy, but not when computing average precision curves. Thus, we perform significance testing for a difference in accuracy on VOC 2007 rather than a difference in mAP. A caveat of this procedure is that it does not consider run-to-run variability when training the models, only variability arising from using a finite sample of images for evaluation.

B.8.2. Results with Standard ResNet

The ResNet-50 ($4\times$) results shown in Table 8 of the text show no clear advantage to the supervised or self-supervised models. With the narrower ResNet-50 architecture, however, supervised learning maintains a clear advantage over self-supervised learning. The supervised ResNet-50 model outperforms the self-supervised model on all datasets with linear evaluation, and most (10 of 12) datasets with fine-tuning. The weaker performance of the ResNet model compared to the ResNet ($4\times$)

2007、DTD 和 Oxford 102 Flowers。其他数据集则从训练集中留出验证子集，用于超参数选择。在验证集上选好超参数后，使用选定参数，在全部训练和验证图像上重新训练，并报告测试准确率。

通过线性分类器迁移。冻结预训练网络，提取特征，训练带 ℓ_2 正则化的多项逻辑回归分类器。使用 L-BFGS 优化 softmax 交叉熵，不做数据增强。预处理采用双三次插值，将短边缩放到224像素，再取 224×224 中心裁剪。在 10^{-6} 到 10^{5} 之间按对数间隔取45个值，搜索 ℓ_2 正则化系数。

通过微调进行迁移学习。我们以预训练权重初始化整个网络，再进行微调。使用带 Nesterov 动量的 SGD，动量0.9，批量大小256，训练20000步。将批归一化统计量的动量设为 $\max(1 - 10/s, 0.9)$，其中 s 是每轮的步数。微调增强只包括随机裁剪、缩放和翻转；不同于预训练，不做颜色增强或模糊。测试时，先将短边缩放到256像素，再取 224×224 中心裁剪。进一步优化增强，尤其在 CIFAR-10/100 上，可能还能提高准确率。我们搜索学习率和权重衰减：在0.0001至0.1之间按对数间隔选择7个学习率，在 10^{-6} 至 10^{-3} 之间按对数间隔选择7个权重衰减值，另加不使用权重衰减的情况。实际使用时，将这些权重衰减值除以学习率。

从随机初始化开始训练。我们采用与微调相同的流程，但训练更久，并调整超参数网格。学习率在0.001至1.0间按对数间隔取7个值，权重衰减在 10^{-5} 至 $10^{-1.5}$ 间取8个值。重要的是，随机初始化基线训练40000步，这足以达到接近最高的准确率，如 Kornblith 等（2019）的图8所示。

Birdsnap 上，各方法没有统计显著差异；Food-101、Stanford Cars 和 FGVC Aircraft 上，微调相较随机初始化训练只有小幅优势；其余8个数据集上，预训练有明确优势。

监督基线。我们比较了架构完全相同、在 ImageNet 上以标准交叉熵训练的 ResNet。这些模型采用与自监督模型相同的增强，包括裁剪、强颜色增强和模糊，也训练1000轮。虽然较强增强和更长训练没有改善 ImageNet 准确率，但在部分迁移数据集的线性评估中，它们明显优于只训练90轮、使用常规增强的监督基线。监督 ResNet-50 在 ImageNet 上达到76.3% top-1，对应自监督模型为69.3%；监督 ResNet-50（$4\times$）为78.3%，对应自监督模型为76.5%。

统计显著性检验。我们用置换检验判断模型差异是否显著。给定两个模型的预测，对每个样本随机交换两者预测，再计算准确率差异；重复生成100000个零假设分布样本，计算其中比实际预测差异更极端的比例。对于 top-1 准确率，这与精确 McNemar 检验结果相同。零假设下的可交换性，也适用于类别平均准确率，但不适用于平均精度曲线。因此，在 VOC 2007 上，我们检验准确率差异，而非 mAP 差异。需要注意，该检验只考虑有限评估样本造成的波动，不考虑训练运行之间的波动。

B.8.2 标准 ResNet 的结果

正文表8中的 ResNet-50（$4\times$）结果，未显示监督或自监督模型具有明确优势。但使用较窄的标准 ResNet-50 时，监督学习仍明显优于自监督学习：在线性评估的全部数据集及微调的12个数据集中10个，监督模型更好。标准 ResNet 相比 ResNet（$4\times$）较弱的表现，

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
Linear evaluation:												
SimCLR (ours)	68.4	90.6	71.6	37.4	58.8	50.3	50.3	80.5	**74.5**	83.6	90.3	91.2
Supervised	**72.3**	**93.6**	**78.3**	**53.7**	**61.9**	**66.7**	**61.0**	**82.8**	74.9	**91.5**	**94.5**	**94.7**
Fine-tuned:												
SimCLR (ours)	**88.2**	**97.7**	**85.9**	**75.9**	63.5	91.3	**88.1**	84.1	**73.2**	89.2	92.1	97.0
Supervised	**88.3**	**97.5**	**86.4**	**75.8**	**64.3**	**92.1**	86.0	**85.0**	**74.6**	**92.1**	**93.3**	**97.6**
Random init	86.9	95.9	80.2	**76.1**	53.6	91.4	85.9	67.3	64.8	81.5	72.6	92.0

Table B.5. Comparison of transfer learning performance of our self-supervised approach with supervised baselines across 12 natural image datasets, using ImageNet-pretrained ResNet models. See also Figure 8 for results with the ResNet $(4\times)$ architecture.

model may relate to the accuracy gap between the supervised and self-supervised models on ImageNet. The self-supervised ResNet gets 69.3% top-1 accuracy, 6.8% worse than the supervised model in absolute terms, whereas the self-supervised ResNet $(4\times)$ model gets 76.5%, which is only 1.8% worse than the supervised model.

B.9. CIFAR-10

While we focus on using ImageNet as the main dataset for pretraining our unsupervised model, our method also works with other datasets. We demonstrate it by testing on CIFAR-10 as follows.

Setup As our goal is not to optimize CIFAR-10 performance, but rather to provide further confirmation of our observations on ImageNet, we use the same architecture (ResNet-50) for CIFAR-10 experiments. Because CIFAR-10 images are much smaller than ImageNet images, we replace the first 7x7 Conv of stride 2 with 3x3 Conv of stride 1, and also remove the first max pooling operation. For data augmentation, we use the same Inception crop (flip and resize to 32x32) as ImageNet,[15] and color distortion (strength=0.5), leaving out Gaussian blur. We pretrain with learning rate in $\{0.5, 1.0, 1.5\}$, temperature in $\{0.1, 0.5, 1.0\}$, and batch size in $\{256, 512, 1024, 2048, 4096\}$. The rest of the settings (including optimizer, weight decay, etc.) are the same as our ImageNet training.

Our best model trained with batch size 1024 can achieve a linear evaluation accuracy of 94.0%, compared to 95.1% from the supervised baseline using the same architecture and batch size. The best self-supervised model that reports linear evaluation result on CIFAR-10 is AMDIM (Bachman et al., 2019), which achieves 91.2% with a model $25\times$ larger than ours. We note that our model can be improved by incorporating extra data augmentations as well as using a more suitable base network.

Performance under different batch sizes and training steps Figure B.7 shows the linear evaluation performance under different batch sizes and training steps. The results are consistent with our observations on ImageNet, although the largest batch size of 4096 seems to cause a small degradation in performance on CIFAR-10.

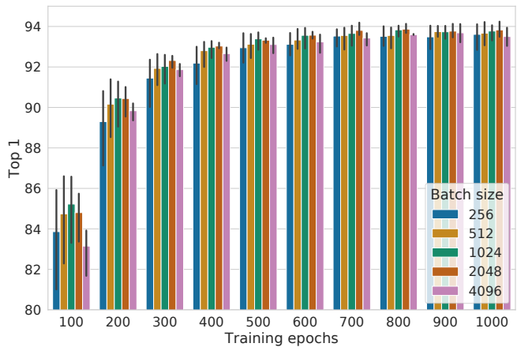


Figure B.7. Linear evaluation of ResNet-50 (with adjusted stem) trained with different batch size and epochs on CIFAR-10 dataset. Each bar is averaged over 3 runs with different learning rates (0.5, 1.0, 1.5) and temperature $\tau = 0.5$. Error bar denotes standard deviation.

[15] It is worth noting that, although CIFAR-10 images are much smaller than ImageNet images and image size does not differ among examples, cropping with resizing is still a very effective augmentation for contrastive learning.

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
线性评估：												
SimCLR（本文）	68.4	90.6	71.6	37.4	58.8	50.3	50.3	80.5	74.5	83.6	90.3	91.2
监督	**72.3**	**93.6**	**78.3**	**53.7**	**61.9**	**66.7**	**61.0**	**82.8**	74.9	**91.5**	**94.5**	**94.7**
微调：												
SimCLR（本文）	88.2	**97.7**	**85.9**	**75.9**	63.5	91.3	**88.1**	84.1	73.2	89.2	92.1	97.0
监督	**88.3**	**97.5**	**86.4**	**75.8**	**64.3**	**92.1**	86.0	**85.0**	**74.6**	**92.1**	**93.3**	**97.6**
随机初始化	86.9	95.9	80.2	**76.1**	53.6	91.4	85.9	67.3	64.8	81.5	72.6	92.0

表B.5：在12个自然图像数据集上，将自监督方法与监督基线的迁移性能比较。均使用 ImageNet 预训练的 ResNet。ResNet（4×）结果另见图8。

可能与 ImageNet 上监督和自监督模型的准确率差距有关。自监督标准 ResNet 的 top-1 为69.3%，比监督模型低6.8个百分点；自监督 ResNet（4×）为76.5%，只低1.8个百分点。

B.9 CIFAR-10

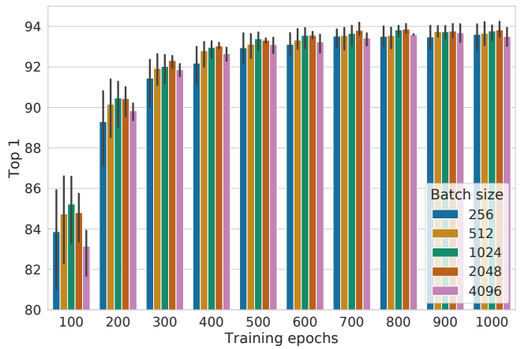
虽然主要使用 ImageNet 预训练无监督模型，我们的方法也适用于其他数据集。下面用 CIFAR-10 实验说明。

设置。我们的目标并非优化 CIFAR-10 性能，而是进一步确认 ImageNet 上的观察，因此在 CIFAR-10 实验中仍使用 ResNet-50。由于 CIFAR-10 图像小得多，我们将首个步幅2的7×7卷积替换为步幅1的3×3卷积，并移除首个最大池化操作。增强使用与 ImageNet 相同的 Inception 裁剪，包括翻转并缩放至32×32，[15] 以及强度为 =0.5 的颜色扰动，但不使用高斯模糊。预训练时搜索学习率 $\{0.5, 1.0, 1.5\}$、温度 $\{0.1, 0.5, 1.0\}$ 和批量大小 $\{256, 512, 1024, 2048, 4096\}$。其余设置，包括优化器与权重衰减，均与 ImageNet 训练相同。

批量1024训练的最佳模型，线性评估达到94.0%，同架构、同批量的监督基线为95.1%。此前报告 CIFAR-10 线性评估的最佳自监督方法 AMDIM（Bachman 等，2019），用一个模型达到91.2%，

比我们的模型大 25× 。加入额外增强，并使用更合适的基础网络，还可能改善我们的模型。

不同批量与训练步数的性能。图B.7展示对应线性评估。结果与 ImageNet 上的观察一致，但最大批量4096似乎导致 CIFAR-10 性能略降。



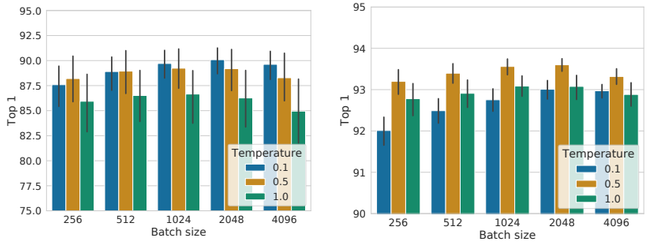
图B.7：调整输入端的 ResNet-50，在 CIFAR-10 上使用不同批量和训练轮数后的线性评估。每根柱对不同学习率和温度的

3 0 5, 1 0, 次运行取平均，误差线表示标准差。

1 5 $\tau = 0.5$

[15] 虽然 CIFAR-10 图像远小于 ImageNet，且各样本尺寸相同，裁剪缩放仍是对比学习非常有效的增强。

Optimal temperature under different batch sizes Figure B.8 shows the linear evaluation of model trained with three different temperatures under various batch sizes. We find that when training to convergence (e.g. training epochs > 300), the optimal temperature in $\{0.1, 0.5, 1.0\}$ is 0.5 and seems consistent regardless of the batch sizes. However, the performance with $\tau = 0.1$ improves as batch size increases, which may suggest a small shift of optimal temperature towards 0.1.



(a) Training epochs ≤ 300 (b) Training epochs > 300

Figure B.8. Linear evaluation of the model (ResNet-50) trained with three temperatures on different batch sizes on CIFAR-10. Each bar is averaged over multiple runs with different learning rates and total train epochs. Error bar denotes standard deviation.

B.10. Tuning For Other Loss Functions

The learning rate that works best for NT-Xent loss may not be a good learning rate for other loss functions. To ensure a fair comparison, we also tune hyperparameters for both margin loss and logistic loss. Specifically, we tune learning rate in $\{0.01, 0.1, 0.3, 0.5, 1.0\}$ for both loss functions. We further tune the margin in $\{0, 0.4, 0.8, 1.6\}$ for margin loss, the temperature in $\{0.1, 0.2, 0.5, 1.0\}$ for logistic loss. For simplicity, we only consider the negatives from one augmentation view (instead of both sides), which slightly impairs performance but ensures fair comparison.

C. Further Comparison to Related Methods

As we have noted in the main text, most individual components of SimCLR have appeared in previous work, and the improved performance is a result of a combination of these design choices. Table C.1 provides a high-level comparison of the design choices of our method with those of previous methods. Compared with previous work, our design choices are generally simpler.

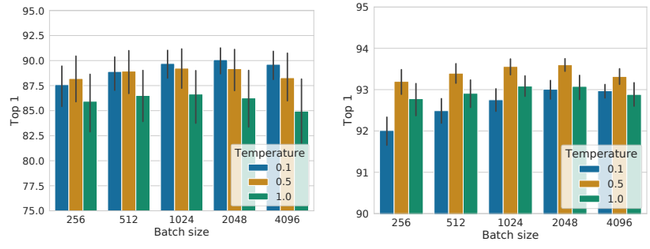
Model	Data Augmentation	Base Encoder	Projection Head	Loss	Batch Size	Train Epochs
CPC v2	Custom	ResNet-161 (modified)	PixelCNN	Xent	$512^{\#}$	$\sim$200
AMDIM	Fast AutoAug.	Custom ResNet	Non-linear MLP	Xent w/ clip,reg	$1008^{\#}$	150
CMC	Fast AutoAug.	ResNet-50 (2×, L+ab)	Linear layer	Xent w/ ℓ_2, τ	156^{*}	280
MoCo	Crop+color	ResNet-50 (4×)	Linear layer	Xent w/ ℓ_2, τ	256^{*}	200
PIRL	Crop+color	ResNet-50 (2×)	Linear layer	Xent w/ ℓ_2, τ	1024^{*}	800
SimCLR	Crop+color+blur	ResNet-50 (4×)	Non-linear MLP	Xent w/ ℓ_2, τ	4096	1000

Table C.1. A high-level comparison of design choices and training setup (for best result on ImageNet) for each method. Note that descriptions provided here are general; even when they match for two methods, formulations and implementations may differ (e.g. for color augmentation). Refer to the original papers for more details. $^{\#}$Examples are split into multiple patches, which enlarges the effective batch size. *A memory bank is employed.

In below, we provide an in-depth comparison of our method to the recently proposed contrastive representation learning methods:

- DIM/AMDIM (Hjelm et al., 2018; Bachman et al., 2019) achieve global-to-local/local-to-neighbor prediction by predicting the middle layer of ConvNet. The ConvNet is a ResNet that has bewen modified to place significant constraints on the receptive fields of the network (e.g. replacing many 3x3 Convs with 1x1 Convs). In our framework, we decouple the prediction task and encoder architecture, by random cropping (with resizing) and using the final

不同批量下的最佳温度。图B.8展示三种温度、不同批量训练模型的线性评估。我们发现，训练至收敛时，例如训练轮数 >300，$\{0.1, 0.5, 1.0\}$ 中的最佳温度为0.5，似乎不随批量改变。但 $\tau = 0.1$ 的表现随批量增大而改善，可能说明最佳温度稍微向0.1移动。



(a) Training epochs ≤ 300 (b) Training epochs > 300

图B.8：CIFAR-10 上，ResNet-50 使用三种温度及不同批量训练后的线性评估。每根柱对不同学习率和总训练轮数的多次运行取平均，误差线为标准差。

B.10 其他损失函数的参数优化

对 NT-Xent 最合适的学习率，未必适合其他损失。为公平比较，我们也优化间隔损失与逻辑损失的超参数。两者都在 $\{0.01, 0.1, 0.3, 0.5, 1.0\}$ 中搜索学习率；间隔损失另在 $\{0, 0.4, 0.8, 1.6\}$ 中搜索间隔，逻辑损失另在 $\{0.1, 0.2, 0.5, 1.0\}$ 中搜索温度。为简化，只考虑来自一种增强视图的负样本，而非两边都使用。这略微降低性能，但保证公平比较。

C 与相关方法进一步比较

如正文所述，SimCLR 的多数单独组件都出现于以往工作中，性能提升来自这些选择的组合。表C.1总体比较了本方法与此前方法的设计。我们的设计通常更简单。

模型	数据增强	基础编码器	投影头	Loss	批量大小	训练轮数
CPC v2	自定义	ResNet-161（修改版）	PixelCNN	Xent	$512^{\#}$	$\sim$200
AMDIM	Fast AutoAug.	自定义 ResNet	非线性 MLP	Xent，使用裁剪和正则化	$1008^{\#}$	150
CMC	Fast AutoAug.	ResNet-50 (2×, L+ab)	线性层	Xent，使用 ℓ_2、τ	156^{*}	280
MoCo	裁剪+颜色	ResNet-50 (4×)	线性层	Xent，使用 ℓ_2、τ	256^{*}	200
PIRL	裁剪+颜色	ResNet-50 (2×)	线性层	Xent，使用 ℓ_2、τ	1024^{*}	800
SimCLR	裁剪+颜色+模糊	ResNet-50 (4×)	非线性 MLP	Xent，使用 ℓ_2、τ	4096	1000

表C.1：各方法设计选择与训练设置的总体比较，以 ImageNet 最佳结果为准。这里的描述是概括性的；即使两个方法的描述相同，其形式和实现也可能不同，例如颜色增强。详情请参阅原论文。#表示样本被分成多个图像块，使有效批量增大。 *表示使用记忆库。

下面将本方法与近期对比式表征学习方法深入比较：

- DIM/AMDIM（Hjelm 等，2018；Bachman 等，2019）通过预测卷积网络中间层，实现全局到局部、局部到邻域预测。其 ResNet 对感受野施加明显限制，例如把许多3×3卷积替换为1×1卷积。我们通过随机裁剪缩放和使用最终

representations of two augmented views for prediction, so we can use standard and more powerful ResNets. Our NT-Xent loss function leverages normalization and temperature to restrict the range of similarity scores, whereas they use a tanh function with regularization. We use a simpler data augmentation policy, while they use FastAutoAugment for their best result.

- CPC v1 and v2 (Oord et al., 2018; Hénaff et al., 2019) define the context prediction task using a deterministic strategy to split examples into patches, and a context aggregation network (a PixelCNN) to aggregate these patches. The base encoder network sees only patches, which are considerably smaller than the original image. We decouple the prediction task and the encoder architecture, so we do not require a context aggregation network, and our encoder can look at the images of wider spectrum of resolutions. In addition, we use the NT-Xent loss function, which leverages normalization and temperature, whereas they use an unnormalized cross-entropy-based objective. We use simpler data augmentation.

- InstDisc, MoCo, PIRL (Wu et al., 2018; He et al., 2019; Misra & van der Maaten, 2019) generalize the Exemplar approach originally proposed by Dosovitskiy et al. (2014) and leverage an explicit memory bank. We do not use a memory bank; we find that, with a larger batch size, in-batch negative example sampling suffices. We also utilize a nonlinear projection head, and use the representation before the projection head. Although we use similar types of augmentations (e.g., random crop and color distortion), we expect specific parameters may be different.

- CMC (Tian et al., 2019) uses a separated network for each view, while we simply use a single network shared for all randomly augmented views. The data augmentation, projection head and loss function are also different. We use larger batch size instead of a memory bank.

- Whereas Ye et al. (2019) maximize similarity between augmented and unaugmented copies of the same image, we apply data augmentation symmetrically to both branches of our framework (Figure 2). We also apply a nonlinear projection on the output of base feature network, and use the representation before projection network, whereas Ye et al. (2019) use the linearly projected final hidden vector as the representation. When training with large batch sizes using multiple accelerators, we use global BN to avoid shortcuts that can greatly decrease representation quality.

两种增强视图的表征进行预测，因此可以采用标准且更强的 ResNet。NT-Xent 通过归一化与温度限制相似度分数的范围，而它们用带正则化的 tanh。我们的增强更简单，它们的最佳结果则使用 FastAutoAugment。

• CPC v1v2（Oord 等，2018；Hénaff 等，2019）采用确定性策略将图像分块，再用上下文聚合网络 PixelCNN 聚合图像块，以定义上下文预测任务。基础编码器只能看到远小于原图的图像块。我们将预测任务与编码器架构分离，不需要上下文聚合网络，编码器也能处理更广泛的图像分辨率。此外，我们采用包含归一化和温度的 NT-Xent，而它们使用未归一化的交叉熵目标。我们的增强也更简单。

• InstDisc、MoCo、PIRL（Wu 等，2018；He 等，2019；Misra 与 van der Maaten，2019）推广了 Dosovitskiy 等（2014）最初提出的 Exemplar 方法，并使用显式记忆库。我们不使用记忆库，而发现较大批量时，批内负采样已足够。我们还使用非线性投影头，并采用投影头前的表征。虽然增强种类相似，如随机裁剪、颜色扰动，具体参数预计有所不同。

• CMC（Tian 等，2019）为每种视图使用独立网络；我们让所有随机增强视图共享一个网络。数据增强、投影头及损失也不同。我们使用更大批量，而非记忆库。

• Ye 等（2019）最大化同一图像增强与未增强版本的相似性；我们则对框架两条分支对称增强（图2）。我们还对基础网络输出进行非线性投影，并使用投影前表征；Ye 等（2019）使用线性投影后的最终隐藏向量。多加速器大批量训练时，我们采用全局 BN，避免严重降低表征质量的捷径。